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BEHAVIORAL NEUROSCIENCE 392

Special Topic: Neurobiology of CTE

Brain Injury in Football: Concussions, Repeated Impacts, and CTE

Fall 2026

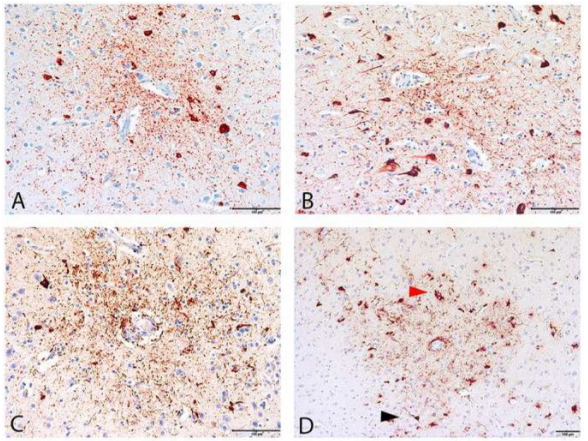
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

CTE is a specific disease process, not a football play

Chronic traumatic encephalopathy, or CTE, is a neurodegenerative disease associated with repeated head impacts. Neurodegeneration means progressive dysfunction and loss of nervous-system cells. CTE is defined by a characteristic microscopic pattern of abnormal tau protein, not by a player's position or one dramatic collision.

The defining lesion contains abnormal phosphorylated tau in neurons around small vessels, especially at the depths of cortical folds. A lesion is a localized area of pathological change. Its cellular and anatomical arrangement distinguishes CTE from simply detecting some tau in an older person's brain.

The Michigan and Northwestern footage provides observable examples of blocking, tackling, and changing head motion. It does not reveal which players have microscopic disease. Movement during a play, accumulated exposure across a career, and the later presence of CTE pathology are different biological measurements.



Bieniek et al. (2021), Fig. 1.



Michigan–Northwestern football plays. FOX Sports (2018).

YouTube source

Play video

Mike Webster's case made microscopic injury visible

Mike Webster played center, an offensive-line position that repeatedly brings a player into close contact with opponents. His postmortem examination became an early scientific report of CTE in a former National Football League player. The diagnosis followed examination of brain tissue rather than analysis of a game photograph.

The 2005 report described tau-positive threads and tangles in neocortical regions, with relatively little gross structural damage. Gross examination means inspecting the brain without a microscope. Molecular and cellular pathology can therefore be present when the organ does not display an obvious large injury.

That individual case raised questions subsequently tested in larger brain-donor studies. It could not establish how frequently CTE occurs among all football players. Its biological contribution was the examination of abnormal proteins and neurons in a documented former player's brain, followed by research using more standardized criteria.

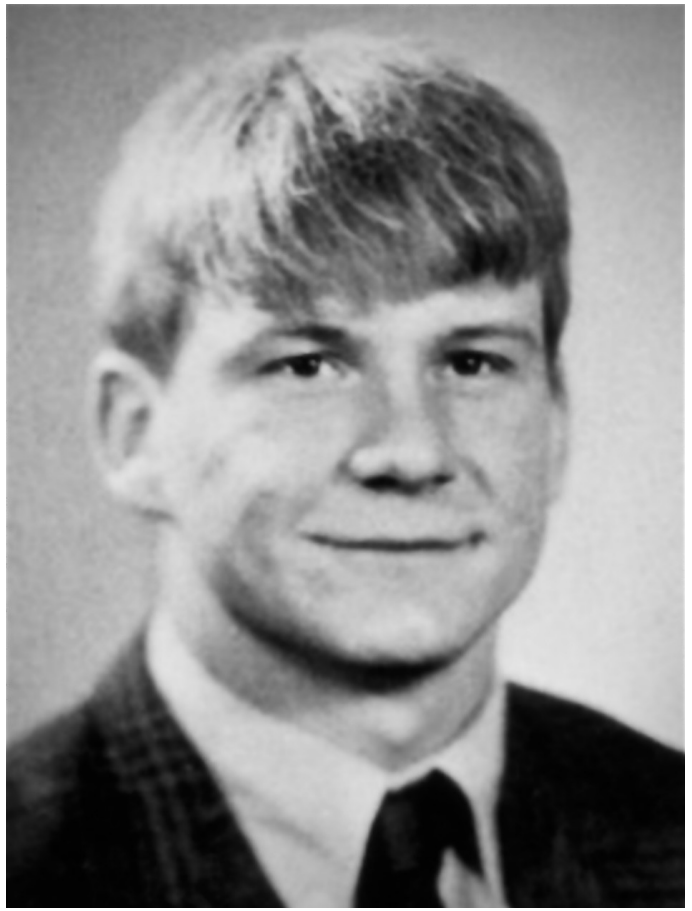


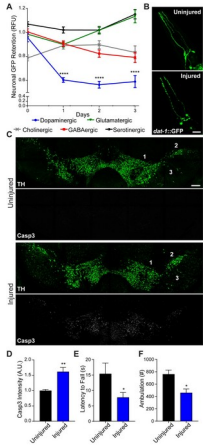
Photo: Mike Webster, 1970. Rhineland High School; public domain.

A collision deforms tissue inside the skull

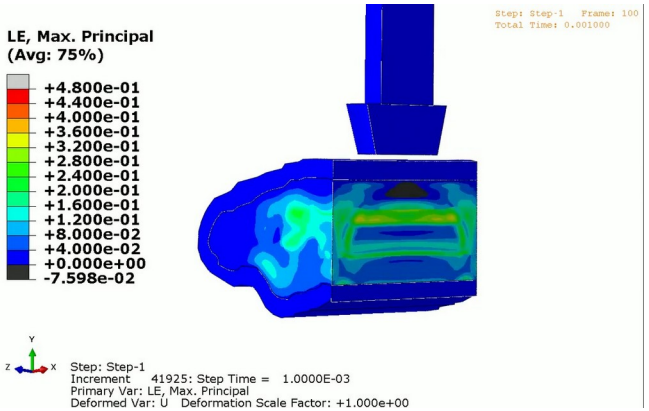
A helmet and skull can reduce some forces reaching the brain, but they do not make the brain mechanically motionless. Translation changes the head's position, while rotation changes its orientation. Acceleration during either movement can deform the soft tissue and the cells embedded within it.

Strain describes a change in a material's shape or length relative to its starting state. Stress describes the internal force acting through an area of material. A short, rapidly applied deformation can disturb axons, membranes, and vessels even without a large visible wound or a skull fracture.

The published impact model follows estimated stress propagating through a mouse brain. Its colors represent calculated mechanical quantities, not a direct recording of tau appearing after impact. The calculation identifies where forces act; cellular experiments and human neuropathology test the subsequent biological changes.



Solano Fonseca et al. (2021), Fig. 1.



Calculated stress after mouse-brain impact. Rene Solano Fonseca (2021).

Play video

Junior Seau's diagnosis required examination of tissue

Junior Seau played linebacker, a position involving pursuit, tackling, and repeated contact. A publicly reported examination by National Institutes of Health researchers identified CTE in his donated brain. The finding concerned a postmortem tissue diagnosis rather than a brain scan performed during his playing career.

Tau is normally part of neuronal cell biology, so a diagnosis requires more than finding that the protein exists. Researchers examine abnormal accumulation, the affected cells, and the distribution across tissue. The diagnostic question concerns a pathological pattern rather than a protein being inherently harmful in every location.

A confirmed lesion does not assign one molecular cause to every symptom or life event experienced by an individual. Different circuits and accompanying diseases can contribute to functioning. The case establishes the need to study damaged neurons and their surroundings without reducing a person's entire history to one tissue finding.



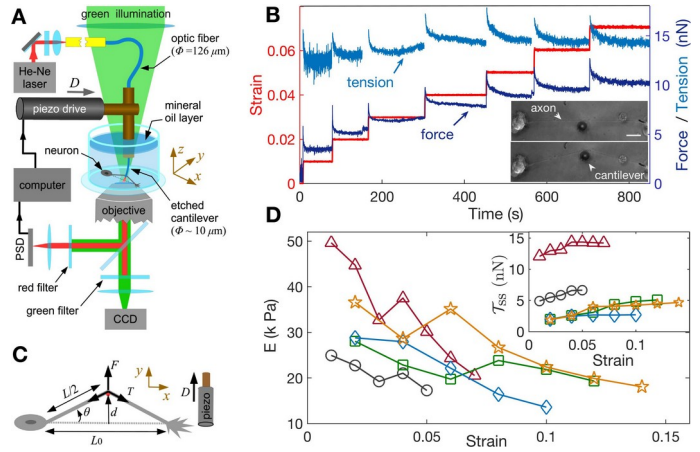
Photo: Junior Seau. Dave Sizer; CC BY 2.0.

Axons must withstand deformation without losing function

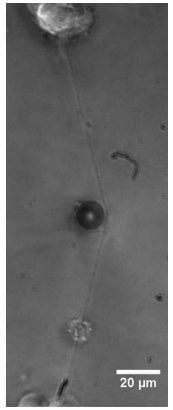
An axon is the long neuronal extension that carries electrical signals toward other cells. Its narrow shape permits communication across distance but also exposes its internal framework to stretching and bending. The membrane and supporting proteins must accommodate ordinary movement while protecting the transport machinery inside.

Microtubules form long protein tracks within axons. Near the membrane, actin rings and spectrin connections provide another supporting scaffold. These components share mechanical loads, so resistance to deformation does not depend solely on the strength of the microtubules at the center.

The original laboratory recording follows a cultured axon as a force apparatus applies successive stretches. Each step changes length while the apparatus measures mechanical behavior. This is a controlled cellular preparation, distinct from directly measuring the forces experienced by a particular football player's axons.



Dubey et al. (2020), Fig. 2.



Cultured axon during controlled stretch. Sushil Dubey (2020).

Play video

Ken Stabler's case separates a player from a biomarker

Ken Stabler played quarterback and later became another publicly reported case of postmortem CTE. His position differed from those of Webster and Seau, as did the pattern of actions required during football. Position identifies an exposure context; it does not independently establish a tissue diagnosis.

CTE pathology is assessed in the brain after death, while symptoms are assessed in a living person. These observations occur at different times and through different methods. A molecular marker intended for use during life must be tested against the pathology it is supposed to identify.

A useful marker would distinguish CTE-related processes from other causes of memory difficulty or behavioral change. Detecting a general injury signal is not sufficient to identify the specific tau lesion. That distinction becomes important when interpreting blood proteins, imaging results, and individual player stories.



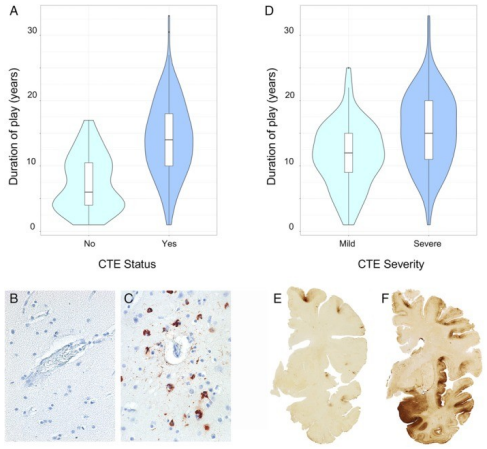
Photo: Ken Stabler, 2007; public domain.

Repeated contact can occur without concussion

A concussion is a traumatic brain injury that produces an acute disturbance of brain function. Repeated head impacts include both recognized concussions and impacts that do not produce an obvious acute syndrome. Absence of a concussion diagnosis does not mean that no mechanical exposure occurred.

The historical Princeton–Yale recording contains ordinary football plays in which bodies repeatedly meet at the line and during tackles. It documents actions and equipment from its own era. It does not provide the acceleration measurements needed to reconstruct each participant's neuronal strain.

Cumulative exposure depends on how often contact occurs, its intensity, and how it is distributed over time. A career's concussion count captures only part of that history. Research therefore examines repeated-impact exposure separately from the number of events remembered or medically labeled as concussions.



Mez et al. (2019), Fig. 2.



Princeton–Yale football, 1903. Edison/Library of Congress (1903).

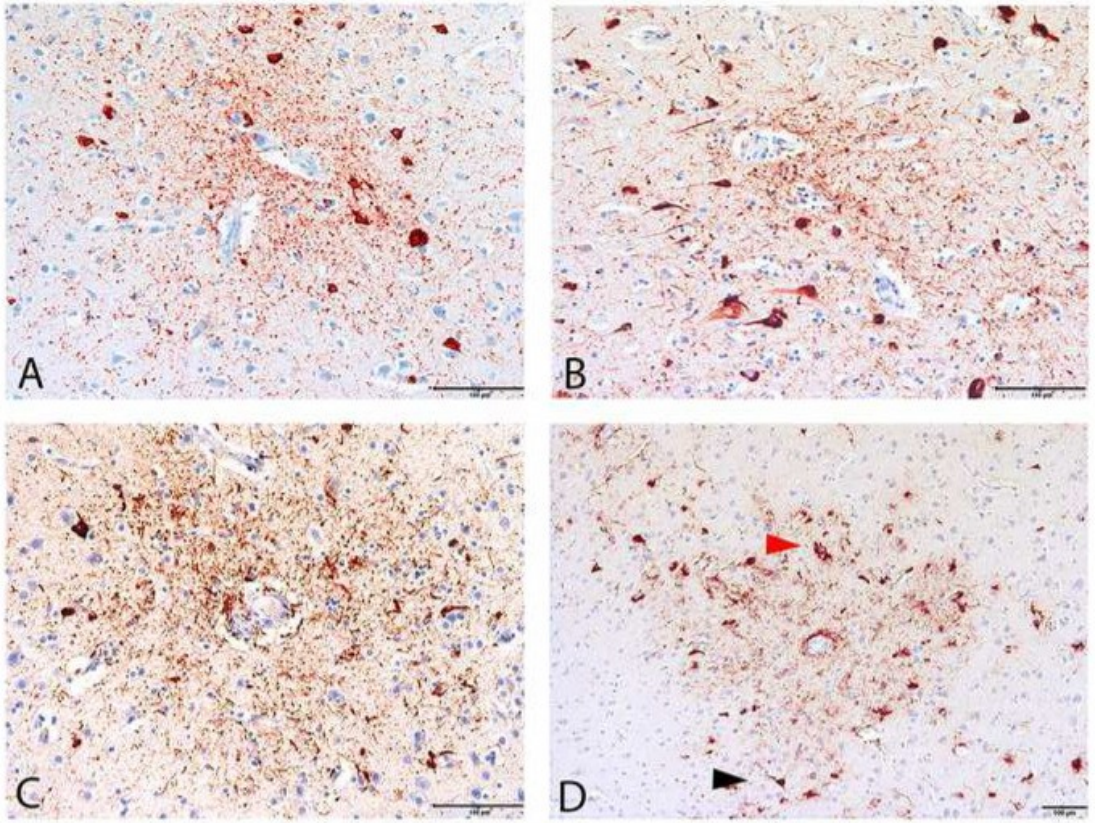
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The defining CTE lesion occupies a particular location

A neuropathology consensus panel examined how reliably investigators could identify CTE among different tauopathies. Tauopathies are diseases involving abnormal tau accumulation. Standardized criteria required the investigators to compare the cells and locations containing the protein rather than diagnose from exposure history alone.

The panel refined the defining lesion to abnormal phosphorylated tau in neurons around small blood vessels at a cortical sulcal depth. A sulcus is a groove between cortical ridges. An irregular distribution around vessels differs from a uniform stain across all layers or every part of the cortex.

Astrocytic tau can accompany this lesion but is not sufficient by itself for the diagnosis. Astrocytes are support cells with roles in metabolism and the vascular interface. Requiring the neuronal component helps distinguish CTE from age-related astrocytic tau patterns that can also involve surface or perivascular regions.



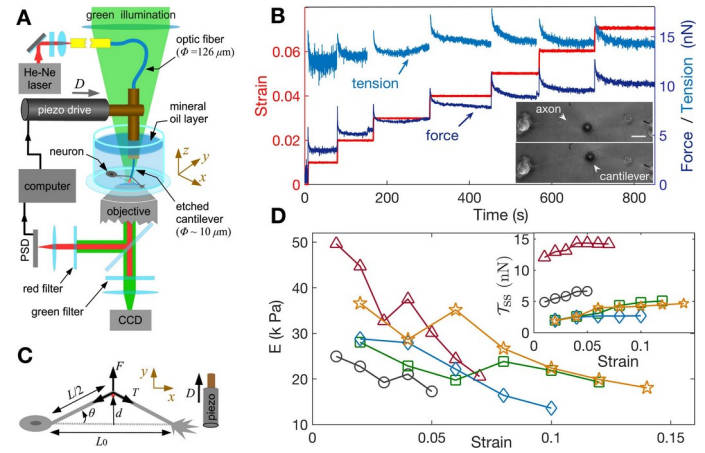
Bieniek et al. (2021), Fig. 1.

An axon can recover after a limited stretch

Elastic deformation reverses when the applied force is removed. Plastic deformation leaves a lasting change after unloading. An axon's response depends on the size and rate of the stretch and on how its supporting structures distribute the load, rather than every change in length causing immediate rupture.

In the original release experiment, the axon shortens toward its earlier length after detachment from the stretching apparatus. This rapid recovery demonstrates a reversible response under the tested conditions. It does not establish that an axon will recover equally well after every faster or larger deformation.

Spectrin contains repeated domains that can unfold under force and thereby release additional length. The study proposed this mechanism as part of a tension buffer. Reversible protein-level changes can absorb load before more damaging alterations affect the membrane, transport tracks, or continuity of the axon.



Dubey et al. (2020), Fig. 2.



Elastic recovery of a stretched axon. Sushil Dubey (2020).

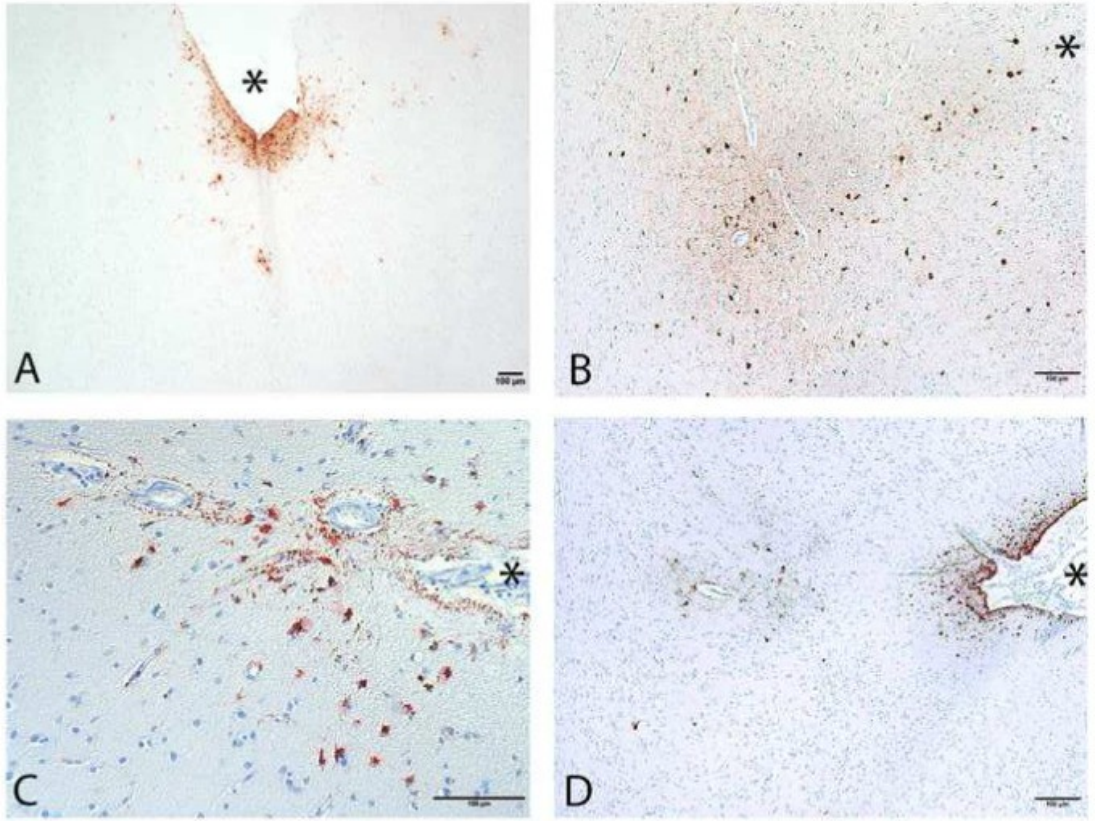
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Neuronal tau distinguishes CTE from some aging patterns

The consensus investigators compared CTE with other conditions in which phosphorylated tau appears. A stain reports a molecular feature, but interpretation also requires recognizing the cell that contains it. Neurons and astrocytes have different shapes, locations, and biological roles within the same tissue section.

Age-related tau astrogliopathy can produce prominent astrocytic staining near the brain surface and around vessels. That finding alone does not meet the neuronal lesion requirement for CTE. The diagnostic boundary therefore rests partly on cellular identity, not merely the depth of a stained region.

The panel also proposed low- and high-severity categories to improve reproducibility. These categories summarize the distribution of pathology across sampled regions. They do not provide a clock that identifies exactly when an individual lesion first formed or the number of impacts that produced it.



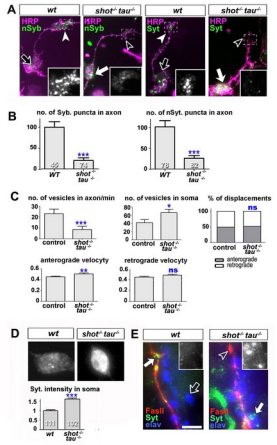
Bieniek et al. (2021), Fig. 2.

Normal tau helps organize neuronal transport

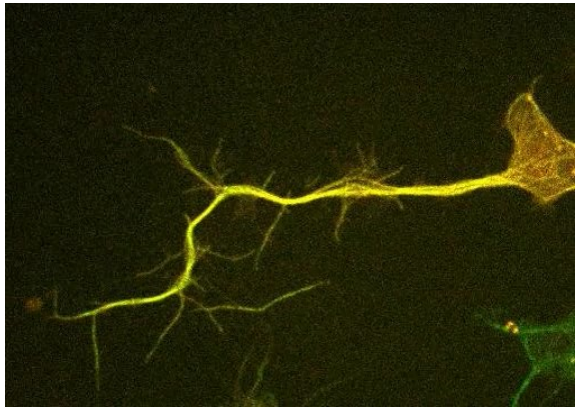
Tau associates with microtubules, the tracks used by motor proteins to move cellular cargo. Motors use chemical energy to carry vesicles and other materials along these tracks. An axon depends on this delivery system because its distant terminals cannot obtain every required component directly from the cell body.

The original fly-neuron recording follows tagged tau together with another microtubule-associated protein. Their overlapping distributions identify tau along the neuronal framework. This observation concerns normal cellular organization rather than demonstrating a CTE lesion or proving that any visible tau is an aggregate.

Changes in tau abundance, binding, or location can affect how the framework supports communication. A protein with useful normal functions can become harmful through altered regulation and accumulation. Loss of normal support and gain of toxic behavior must therefore be examined as separate processes.



Voelzmann et al. (2016), Fig. 4.



Normal tau and microtubules in fly neurons. Andre Voelzmann (2016).

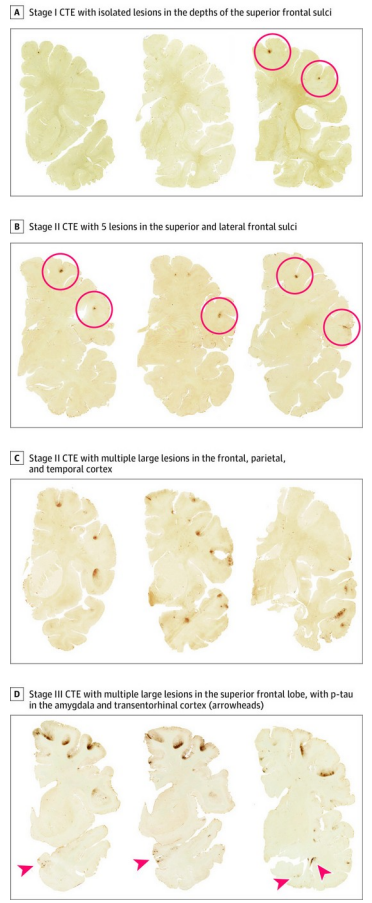
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CTE pathology can occur in young exposed athletes

A brain-donor study examined young contact-sport athletes who had died before age thirty. Some donated brains met CTE neuropathology criteria, usually with relatively limited pathology. The finding established that the characteristic lesion is not restricted to people who survive into advanced old age.

Other abnormalities included changes near vessels and in white matter, the tissue containing many long-range axons. These findings broadened the biological assessment beyond counting tau tangles alone. The study also examined clinical histories reported by people who knew the donors.

Symptoms were common in this selected donor series and were not specific enough to establish CTE pathology during life. The donated group was not a random sample of young athletes. Its results describe tissue and reported histories in the examined donors, rather than the prevalence of CTE among all young football players.



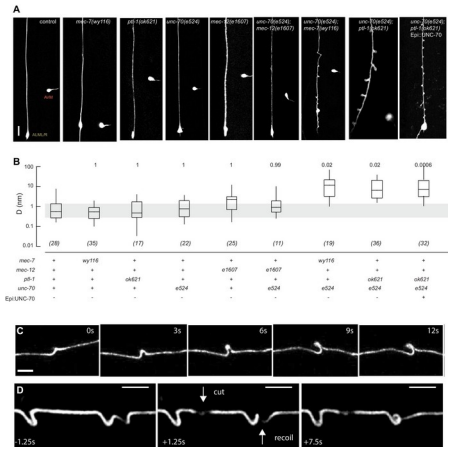
McKee et al. (2023), Fig. 2.

Tau and spectrin cooperate in mechanical protection

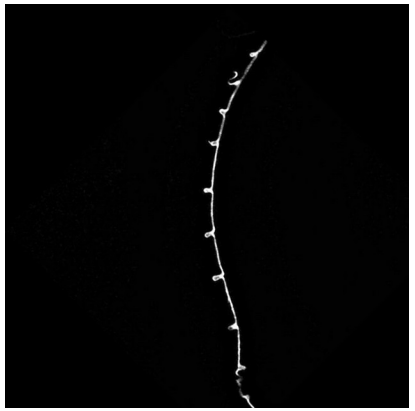
Researchers used the nematode Caenorhabditis elegans to examine how the axonal skeleton resists movement-related damage. They altered spectrin and PTL-1, a tau-related protein, and imaged sensory neurons. The preparation allowed the effect of weakened structural components to be observed in a moving animal.

Combined defects produced abnormal twisting and coiling of axons. The original recording follows a labeled sensory neuron in the double-mutant background. Its shape changes concern a mechanically vulnerable axon, not the formation of the perivascular neuronal tau lesion used to diagnose human CTE.

The study combined live imaging with mechanical tests and structural analysis. Its findings indicated that the spectrin scaffold and microtubule-associated machinery work together. Damage therefore depended on the interaction between cellular architecture and applied movement, rather than one supporting protein acting in isolation.



Krieg et al. (2017), Fig. 1.



Spectrin/tau-mutant worm axon. Michael Krieg (2017).

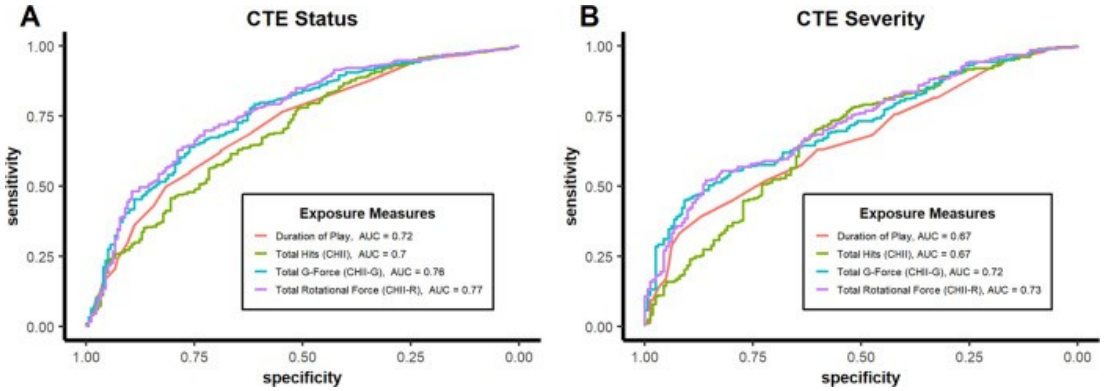
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Impact intensity adds to concussion counts

Daneshvar and colleagues combined published helmet-sensor measurements with football position and level of play. They constructed an exposure matrix that estimated the impacts typically associated with those playing conditions. The estimates were then applied to the histories of a separate group of brain donors.

Estimated cumulative linear and rotational acceleration improved prediction of CTE pathology compared with years of play or impact count alone. Recalled concussion count did not provide the same information. The result implicated accumulated impact intensity in the exposure–pathology relationship.

The exposure estimates were reconstructed from group-level measurements rather than sensors worn by every donor throughout a career. They help compare biological exposure histories but do not provide a personal diagnostic threshold. A predicted exposure burden remains different from observing tau pathology in tissue.



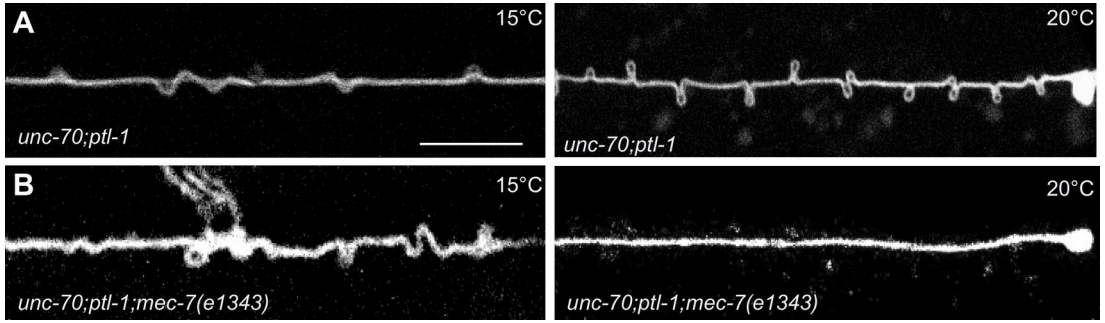
Daneshvar et al. (2023), Fig. 2.

Mechanical tension changes the shape of an axon

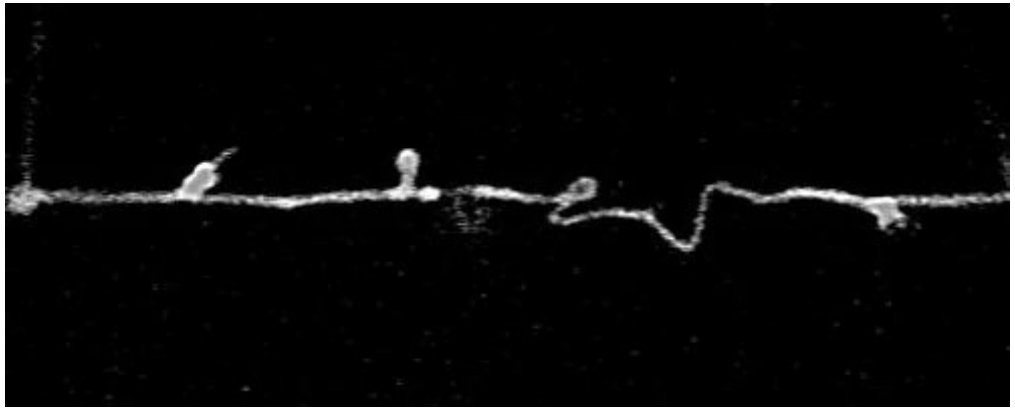
Tension stretches a structure along its length, while torque tends to twist it. An axon's internal framework experiences both kinds of load. The resulting shape depends on the forces and on the resistance supplied by microtubules, membrane-associated proteins, and attachments to surrounding tissue.

In the original worm-neuron experiment, a laser cut changes the mechanical constraints on an axon and is followed by rotation. Releasing a constraint can reveal stored twisting forces. The observed motion therefore provides information about mechanics, not a direct image of a chemical conversion of tau.

A football impact can also produce a mixture of deformation modes, although the scale and preparation differ from laser axotomy. Cellular experiments isolate one contribution at a time. Their mechanical observations help identify which structural failures could precede impaired transport, membrane disturbance, and degeneration.



Krieg et al. (2017), Fig. 10.



Worm axon rotation after laser release. Michael Krieg (2017).

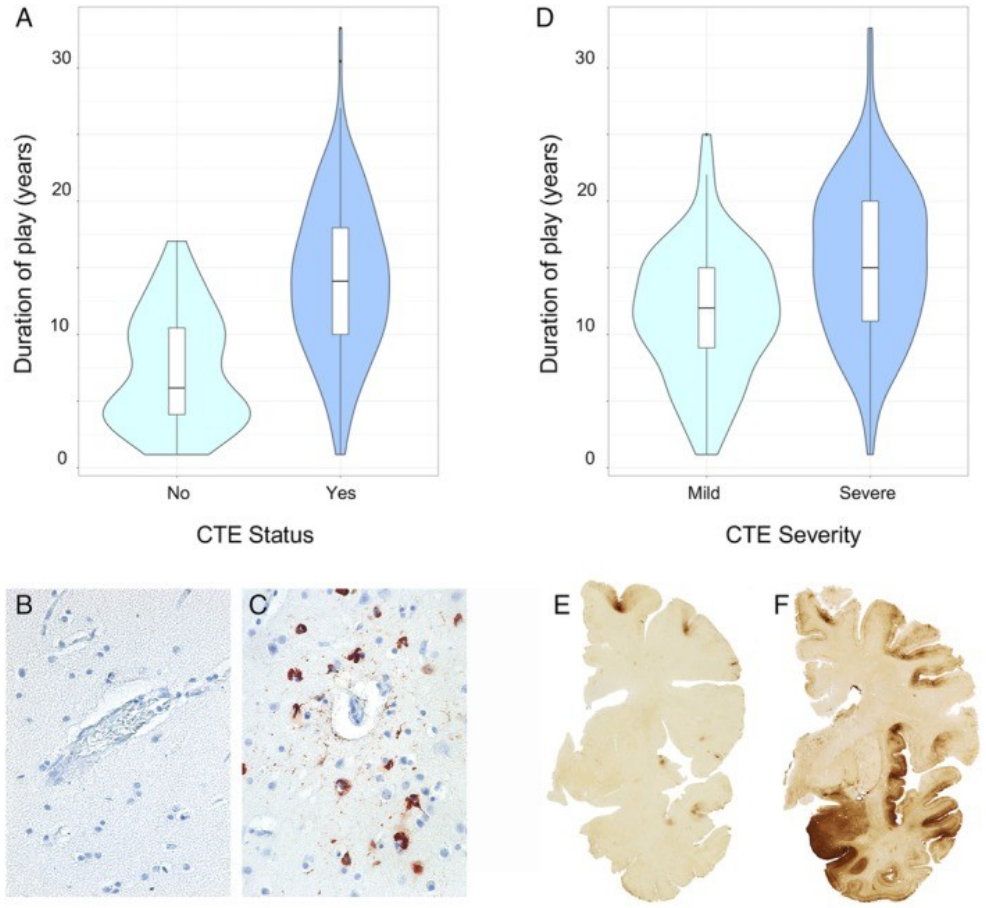
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Longer football exposure tracked CTE pathology in donors

Mez and colleagues compared the duration of American football participation with neuropathology in donated brains. Longer participation was associated with both the presence and severity of CTE. The study therefore tested an exposure gradient rather than treating all people who ever played football as one identical group.

Duration combines many possible opportunities for impact but does not measure every collision directly. Players with similar career lengths can experience different positions, practices, and intensities. The later acceleration-based exposure study tested whether adding information about impact burden improved that description.

The investigators also examined how brain-bank selection could influence the relationship. Donation is not random and can be affected by symptoms or concern about exposure. Analyses addressed the exposure association within the available data, while the percentage of donated brains with CTE remained unsuitable as a population prevalence estimate.



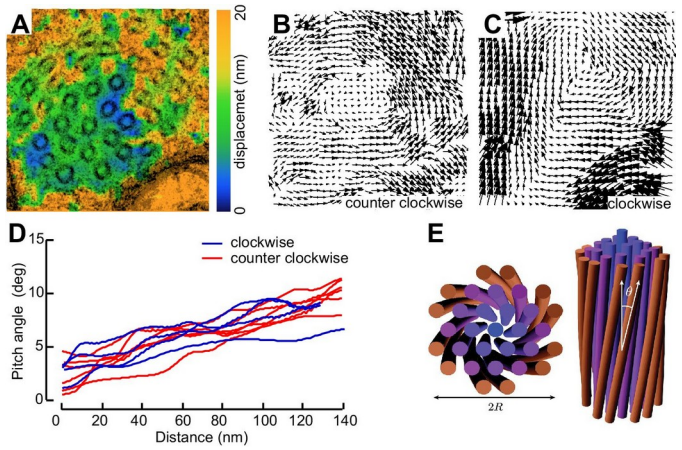
Mez et al. (2019), Fig. 2.

Electron microscopy reveals a twisted internal framework

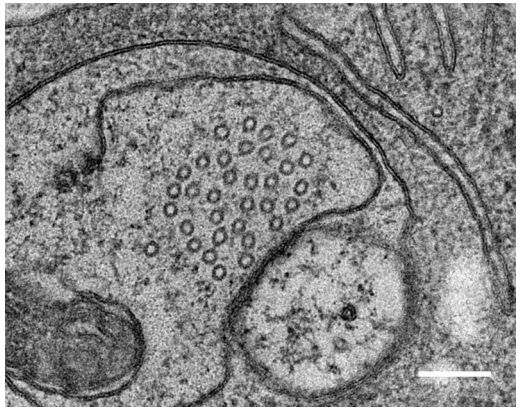
The worm axon study used serial electron microscopy to examine microtubules inside mechanically vulnerable neurons. Electron microscopy resolves structures smaller than those visible through ordinary light imaging. Successive sections allowed the arrangement of internal tracks to be followed through the axon.

The selected original image stack passes through a sensory axon lacking normal tau-related and spectrin function. Microtubules change their winding arrangement along the sampled region. The recording supplies anatomical evidence about the same type of structural instability studied in the live-moving preparation.

The findings supported a combined role for longitudinal tension and twisting forces in maintaining shape. A disorganized framework can interfere with the routes used for cargo movement and electrical continuity. This experiment identifies a cellular mechanical vulnerability, while human CTE additionally requires its specific molecular and anatomical pathology.



Krieg et al. (2017), Fig. 7.



Serial electron images of worm axonal microtubules. Michael Krieg (2017).

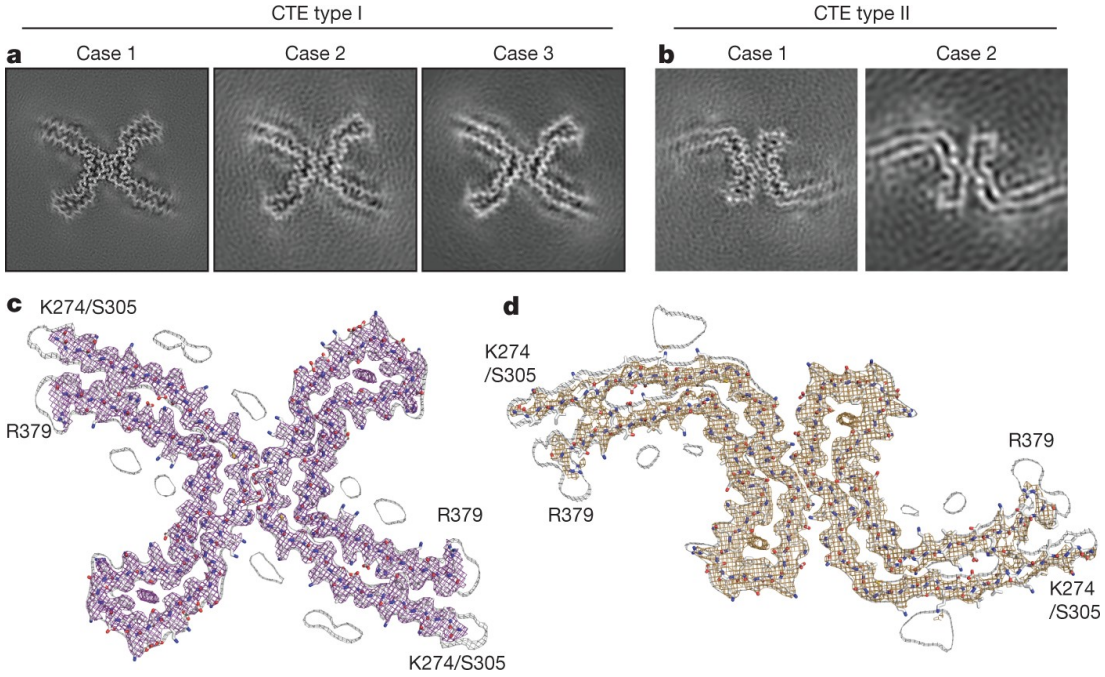
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CTE tau forms an ordered molecular structure

Tau aggregates are not simply random piles of protein. Cryo-electron microscopy of filaments extracted from CTE brains revealed an ordered fold that differs from the common Alzheimer tau fold. Repeated tau molecules pack into a structure that can preserve a particular conformation along the filament.

The CTE fold contains a cavity with density consistent with non-protein material enclosed by tau. The identity and biological role of that material were not established by the structural image alone. The observation nevertheless identifies a physical feature that can influence how the filament is assembled and stabilized.

A molecular fold and a tissue diagnosis operate at different scales. CTE diagnosis combines the affected neurons with their perivascular sulcal distribution. Structural studies identify the arrangement of extracted tau, while neuropathology establishes where the abnormal protein accumulated within the donor's brain.



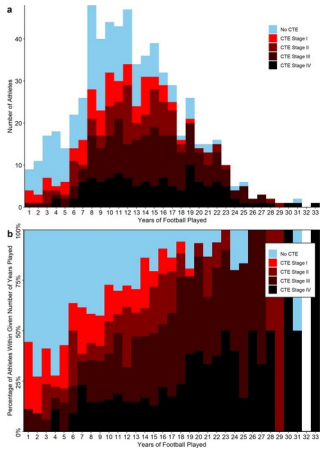
Falcon et al. (2019), Fig. 2.

Football positions create different patterns of contact

Quarterbacks, linemen, and defensive players perform different actions and encounter different patterns of contact. Repeated blocking can generate many exposures without one visually dramatic collision. Open-field pursuit can produce another combination of speed, direction, and contact geometry.

The Oregon–Washington State footage follows pass plays, blocking, pursuit, and tackles in an identified college game. These visible events provide opportunities to distinguish body contact from measured head acceleration. The recording is separate from the donors and sensor datasets used in the CTE exposure studies.

A helmet's external motion does not display the strain of an individual neuron's membrane or the presence of tau filaments. Connecting play to pathology requires several measurements across scales. Football footage establishes the action, sensors quantify motion, and biological studies examine the resulting cellular and tissue changes.



Daneshvar et al. (2023), Fig. 1.



Oregon–Washington State football plays. FOX Sports (2018).

YouTube source

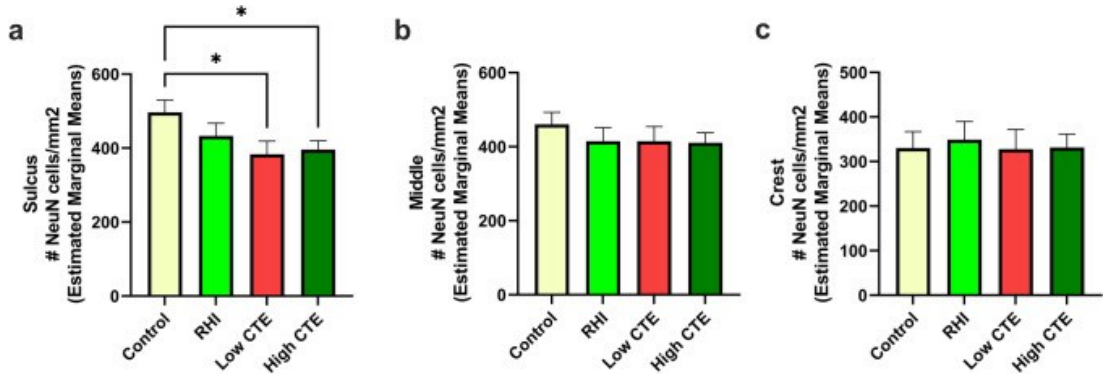
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Sulcal tissue shows a regional pattern of degeneration

A human CTE study measured cortical thickness, neuronal density, and synaptic proteins at the sulcal depth, middle of the fold, and gyral crest. A gyrus is a cortical ridge. Comparing these locations tested whether degeneration follows the same regional preference as the characteristic tau pathology.

CTE tissue showed greater cortical thinning and neuronal loss within the sulcus. Longer contact-sport participation was also associated with these measures. The investigators therefore identified a regional pattern of degeneration rather than inferring neuronal loss solely from the amount of tau staining.

Analyses suggested that the exposure association with neuronal loss was more strongly related to tau pathology than the association with cortical thinning. Different degenerative outcomes need not share one complete mechanism. Measuring cells and tissue thickness separately preserved this distinction within the affected cortical fold.



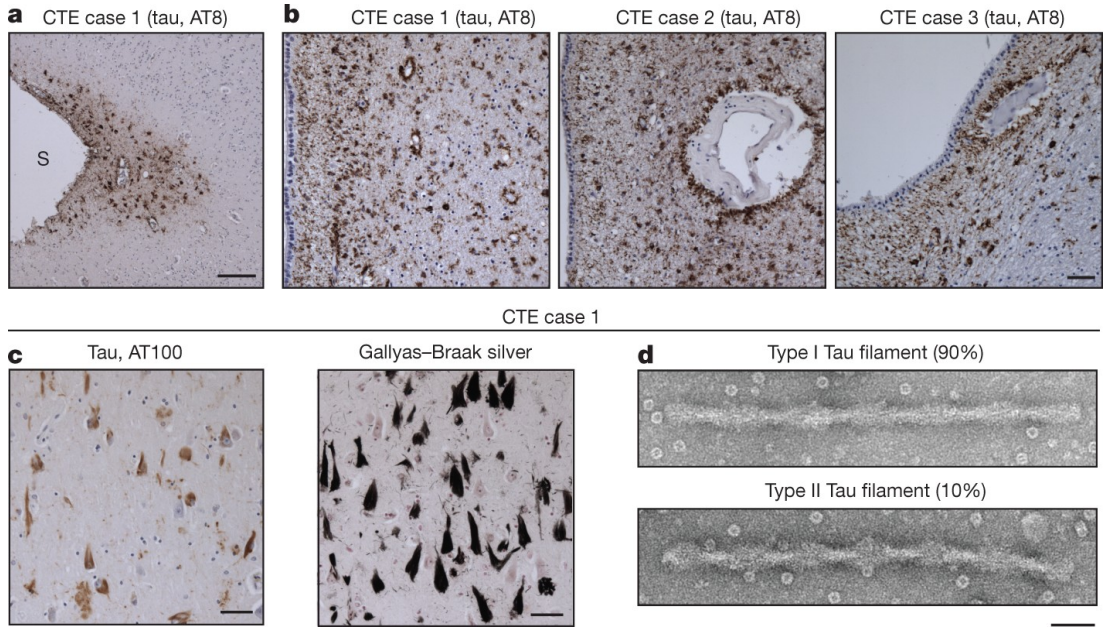
Nicks et al. (2024), Fig. 3.

Phosphorylation changes tau's interactions

Phosphorylation attaches a phosphate group to a protein. Enzymes called kinases add these groups, while phosphatases can remove them. Tau's phosphorylation state changes its interactions with microtubules and other molecules, allowing normal regulation as well as contributing to pathological states.

A phosphorylated tau molecule is not automatically a filament. Phosphorylation, mislocalization, clustering, and ordered aggregation describe different changes. Studying how one change influences another requires measuring those events separately rather than treating every positive tau stain as the same biological process.

In CTE, the combination of abnormal tau accumulation and its neuronal, perivascular distribution defines the lesion. Biochemical experiments address how tau changes state, while tissue examination identifies the pattern of disease. The molecular modification and the anatomical diagnosis therefore provide different kinds of information.



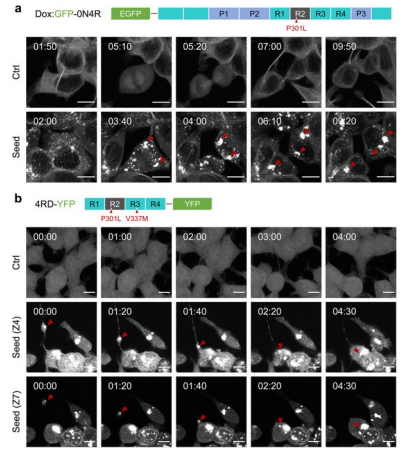
Falcon et al. (2019), Fig. 1.

A tau reporter supplies a controlled cellular baseline

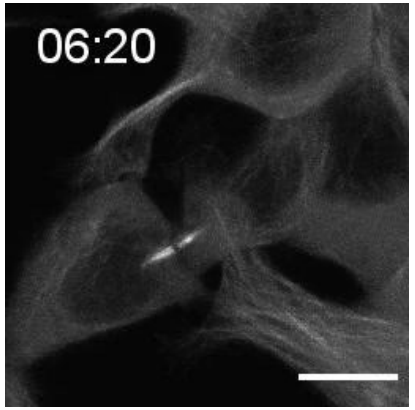
Researchers engineered cells to express tau joined to a fluorescent protein. Fluorescence made the protein's distribution visible while the cells remained alive. The reporter included a four-repeat tau form, allowing changes from dispersed protein to concentrated assemblies to be followed over time.

The selected control recording follows reporter cells exposed to brain material from ordinary mice without the tested seed-competent tau. The signal remains a comparison for the seeded preparations. A control of this kind separates the response to added brain material from the response to pathological tau conformers.

A conformer is a molecule with a particular folded arrangement. The experiment tested whether different tau-containing preparations could impose an altered organization on soluble reporter tau. It examined a defined cell model, rather than identifying CTE in a living football player through fluorescent imaging.



Kang et al. (2021), Fig. 1.



Unseeded control tau reporter cells. Sang-Gyun Kang (2021).

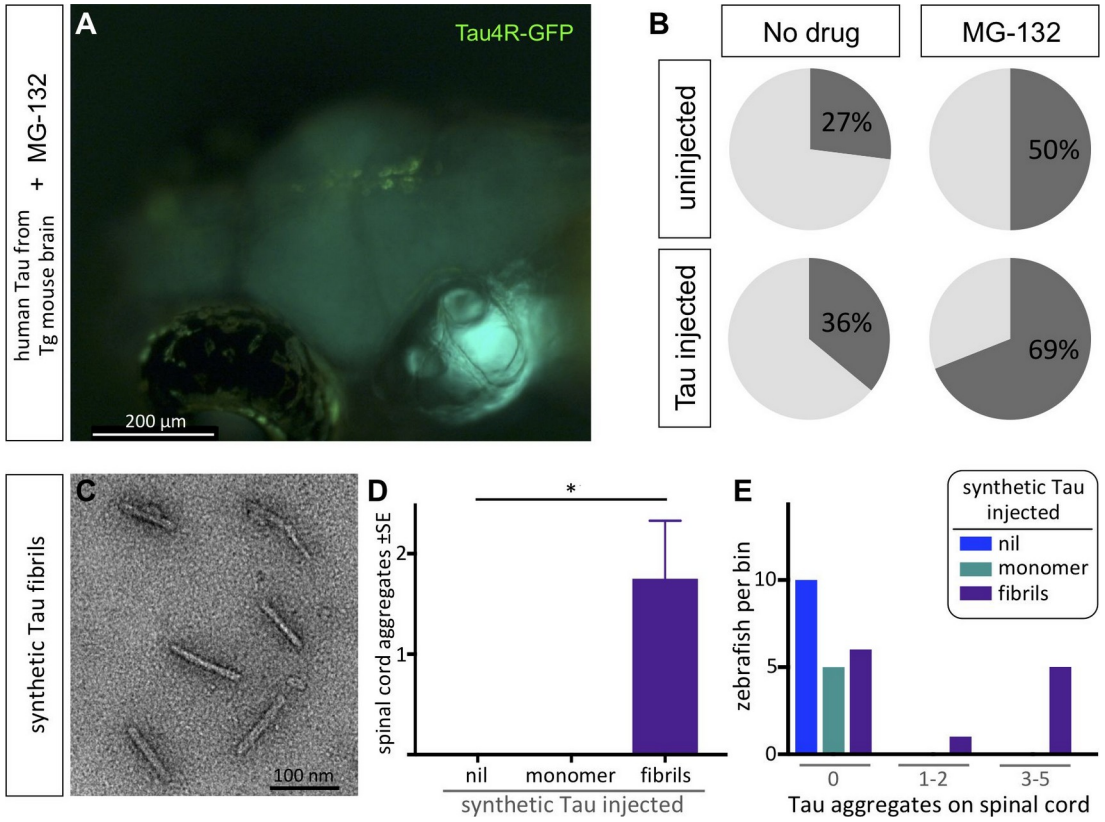
Play video

Seeding describes a change in protein conformation

A protein seed promotes assembly of additional protein molecules into a compatible arrangement. In tau research, this can mean that an existing abnormal assembly helps soluble tau adopt an aggregation-prone state. The process changes molecular organization without requiring a change in the tau gene's DNA sequence.

Seed formation, entry into a cell, escape from internal compartments, and subsequent growth are separate steps. A preparation may contain aggregated protein yet seed poorly if it cannot reach compatible soluble tau. The measured response therefore depends on both the seed and the receiving cell.

Prion-like describes this templated molecular behavior and possible spread between cells. It does not mean that playing near someone with CTE transmits an ordinary contagious illness. Evidence for protein seeding in a laboratory is distinct from evidence about how a disease spreads through a person's brain over years.



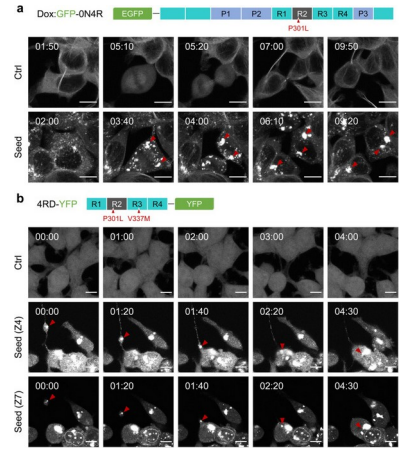
Alyenbaawi et al. (2021), Fig. 2.

Pathological tau redistributes a cell's reporter protein

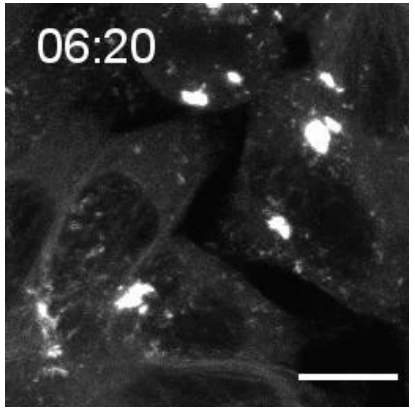
The reporter-cell study added brain fractions from mice expressing P301L tau, a mutation used to model an inherited tauopathy. Those preparations contained seed-competent conformers. The experiment therefore supplied a defined aggregation stimulus rather than relying on the cells to develop pathology spontaneously.

In the original seeded recording, previously dispersed fluorescent tau redistributes into concentrated assemblies. Some assemblies move or fuse, indicating that they are dynamic rather than fixed deposits from the outset. The control and seeded conditions distinguish the effect of the pathological preparation.

The starting material came from a genetic mouse model, not from the football players discussed earlier. Its value is the direct observation of a cellular process that can be tested across tauopathies. A CTE-specific experiment must additionally establish which human-derived conformers and tissue conditions reproduce that response.



Kang et al. (2021), Fig. 1.



Mouse-derived tau seeds in reporter cells. Sang-Gyun Kang (2021).

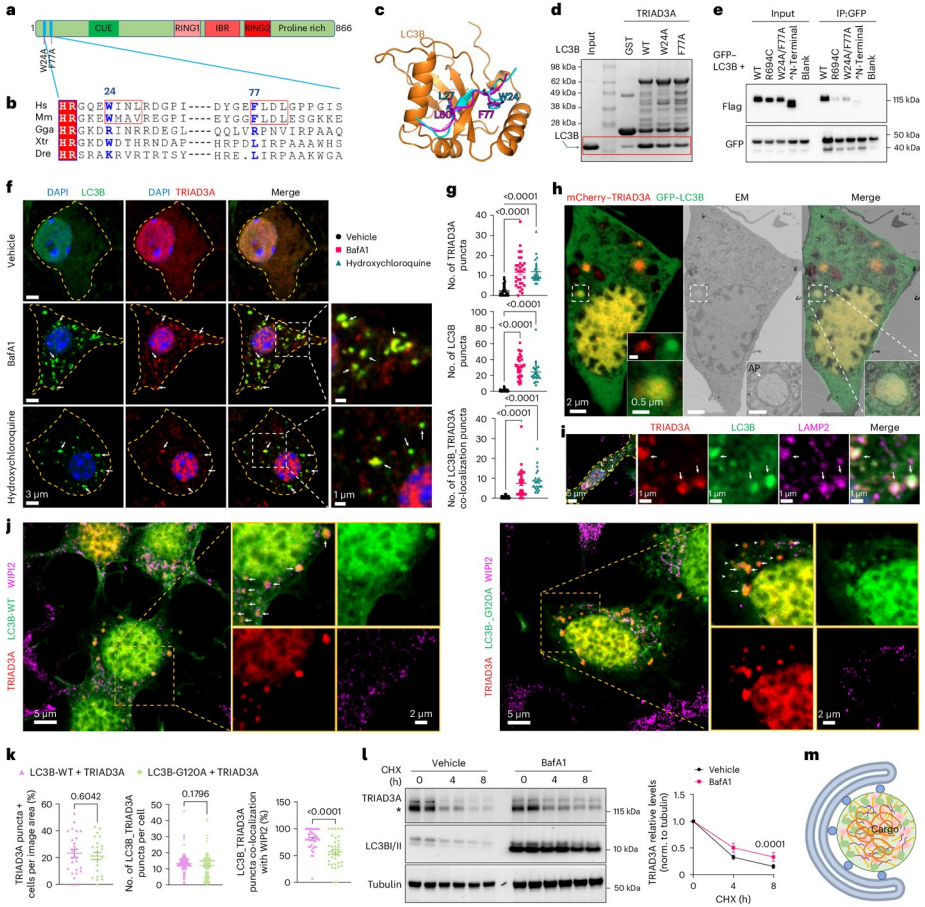
Play video

Protein disposal requires several cellular routes

Cells continually synthesize and remove proteins. The proteasome is a protein-degrading complex that breaks selected proteins into smaller pieces. Ubiquitin tags can help direct a protein toward this machinery, although different ubiquitin arrangements also serve other signaling and trafficking functions.

Autophagy uses membrane-bound compartments to collect material for delivery to lysosomes, organelles containing degradative enzymes. Large assemblies and damaged organelles can require this compartment-based route. Proteasomal breakdown and autophagy therefore handle overlapping but nonidentical kinds of cellular cargo.

Abnormal protein accumulation can reflect increased production, altered folding, impaired removal, or combinations of these processes. Measuring a high abundance of a disposal-related protein does not by itself prove that disposal is working efficiently. The activity and destination of the cargo must also be established.



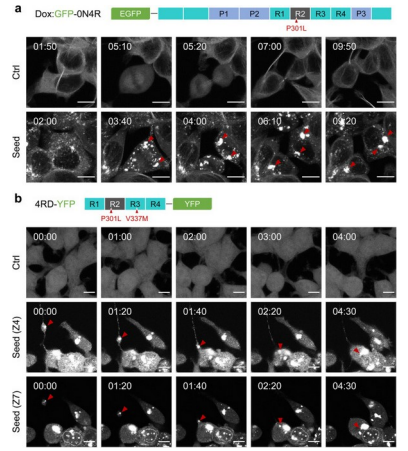
Jiechao et al. (2024), Fig. 4.

Tau assemblies can move through cellular extensions

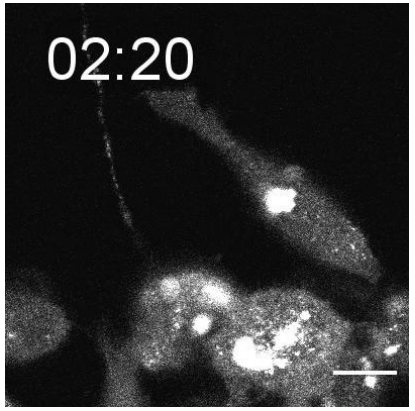
The tau-conformer study followed cells containing a fluorescent four-repeat-domain reporter. After seeding with pathological mouse-brain material, concentrated tau assemblies appeared within cellular processes. Live imaging allowed movement between neighboring cells to be distinguished from the static presence of nearby fluorescent spots.

The selected recording follows a nanotube-like membrane extension at the source's specified focal plane. Fluorescent material moves along the extension between cells. This is one observed transfer route in the reporter preparation, not proof that every CTE lesion spreads through that same anatomical pathway.

The receiving cell's tau provides material that can adopt an altered arrangement after a seed arrives. Transport and templating can therefore occur at different stages. Tracking a moving assembly identifies the route of transfer, while measuring subsequent reporter accumulation tests what happened after it reached another cell.



Kang et al. (2021), Fig. 1.



Tau reporter assemblies and cellular extensions. Sang-Gyun Kang (2021).

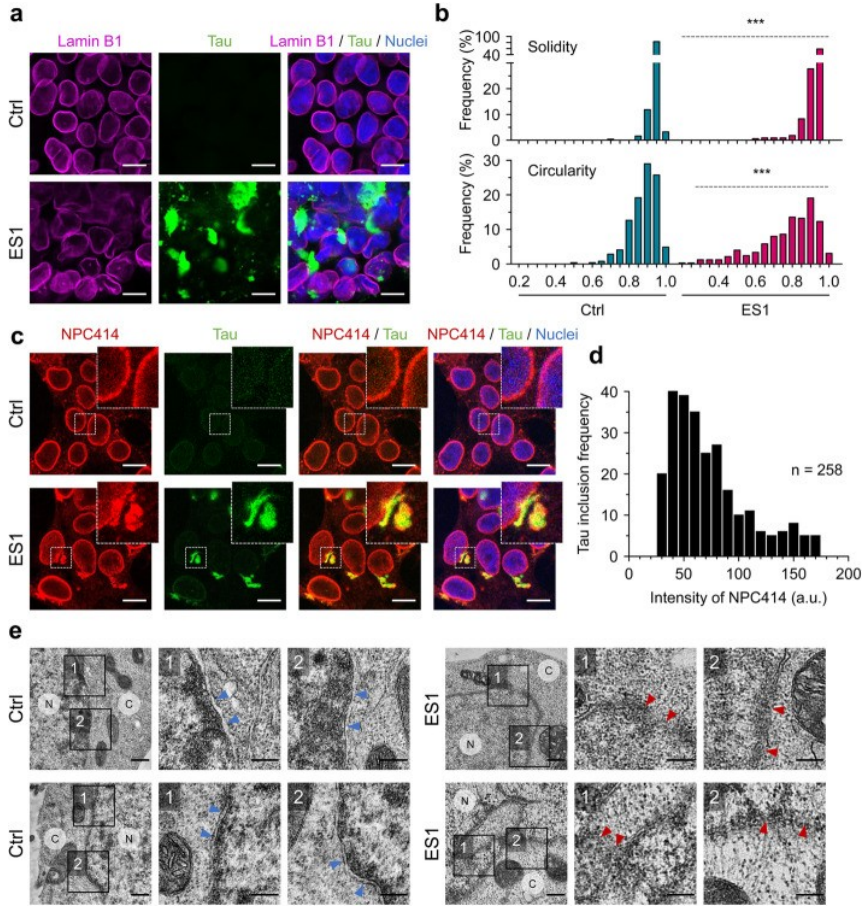
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Nuclear pores regulate exchange with the cytoplasm

The nucleus contains the cell's genome and much of the machinery that produces RNA. Its envelope separates these functions from the surrounding cytoplasm. Nuclear pores are regulated passageways through which selected proteins and RNA move between the two compartments.

Transport proteins recognize signals on their cargo and help it cross the pore. Maintaining the proper distribution of these molecules supports gene regulation and protein production. An altered nuclear boundary can therefore disturb cellular function without immediately destroying the entire nucleus.

Tau assemblies near this boundary raise a specific question about transport rather than merely adding another location to an aggregate list. Researchers can use fluorescent import and export reporters to test exchange. The appearance of an inclusion and a measured transport defect remain separate observations within that experiment.



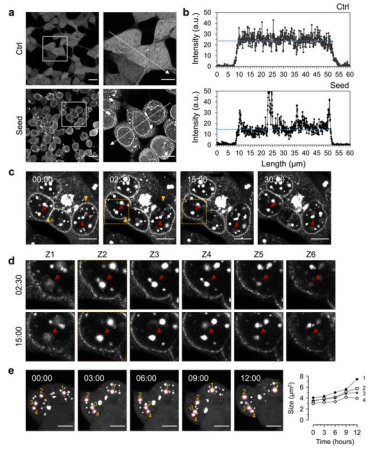
Kang et al. (2021), Fig. 5.

Some nuclear-boundary tau assemblies remain dynamic

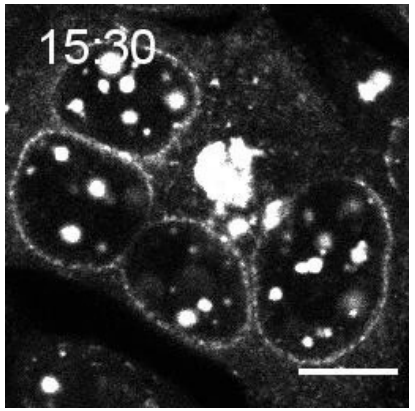
The seeded-cell study found fluorescent tau assemblies near the nuclear envelope. Researchers tested whether molecules could still exchange within these structures using fluorescence recovery after photobleaching, or FRAP. A brief light exposure darkens the label in a selected region without removing the protein itself.

In the original recording, fluorescence recovers within an assembly and droplet-like structures fuse. Recovery indicates movement of labeled molecules back into the bleached region. The observation therefore distinguishes a dynamic condensed state from a completely immobile mass under those experimental conditions.

Larger tau-containing assemblies also developed features associated with more ordered aggregation. A cell can consequently contain structures with different physical properties during the same pathological process. The transition between these states offers a more detailed description than treating all visible inclusions as identical tangles.



Kang et al. (2021), Fig. 2.



Tau exchange at a nuclear boundary. Sang-Gyun Kang (2021).

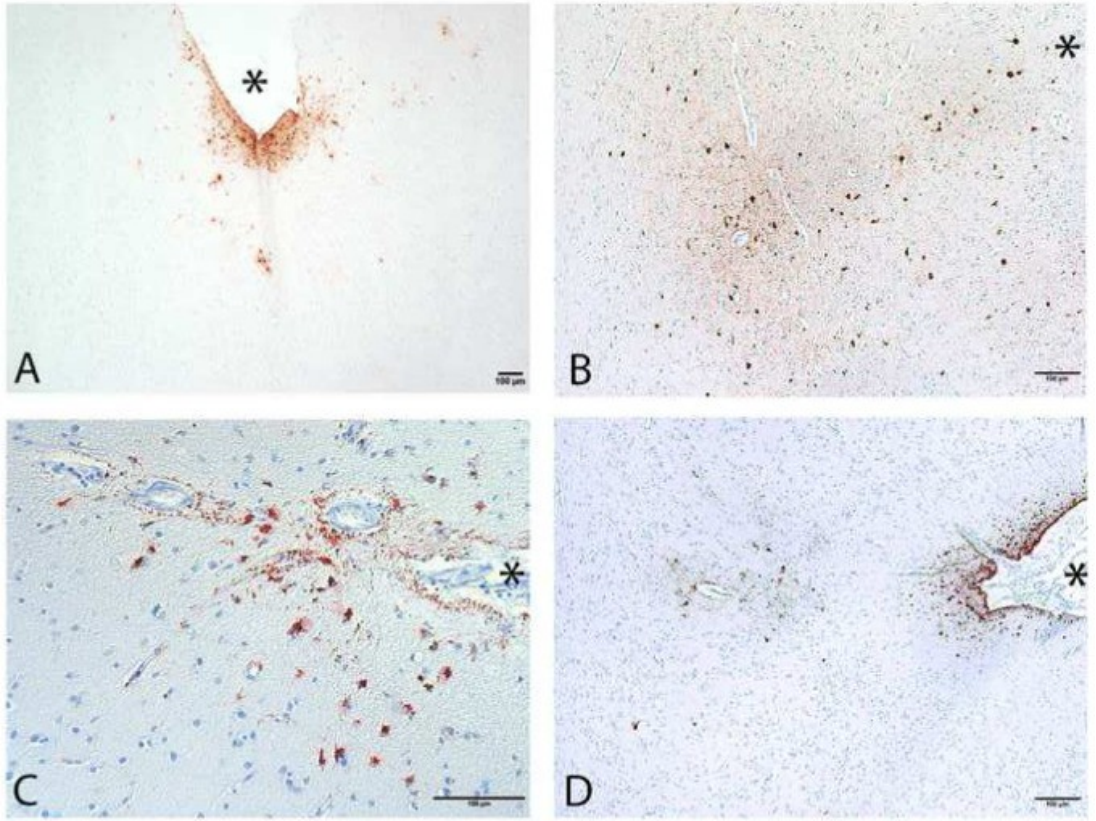
Play video

A tau stain detects a particular molecular feature

An antibody recognizes a molecular feature called an epitope. Some tau antibodies recognize phosphorylation at particular amino-acid positions, while others recognize different parts or states of the protein. Two stains can therefore provide different pictures of the same tissue because they detect different features.

A stronger stain can reflect more protein, a change in modification, or greater accessibility of the recognized feature. It does not directly measure the firing of the stained neuron. Structural, biochemical, and physiological methods are needed to determine what the accumulated protein is doing.

In CTE diagnosis, molecular detection is interpreted together with cellular identity and tissue position. Neuronal staining around a vessel at a sulcal depth carries different information from a diffuse signal in a culture dish. The assay's target must therefore be identified before its result is assigned a disease meaning.



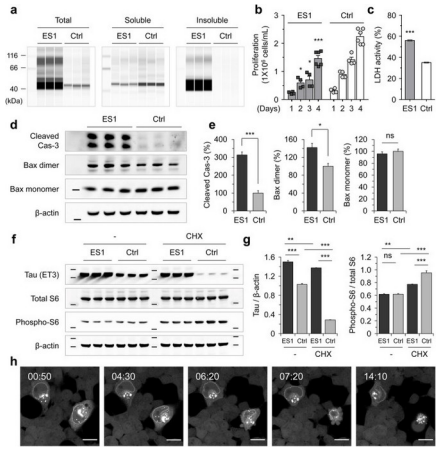
Bieniek et al. (2021), Fig. 2.

Seeded tau assemblies can accompany cellular failure

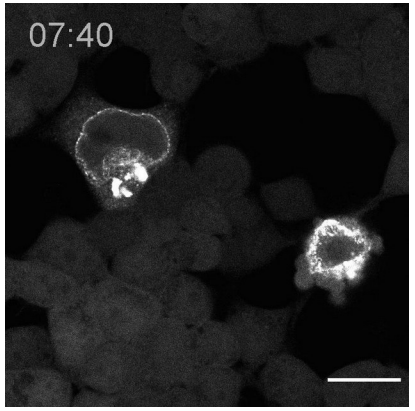
The reporter-cell investigators established a line containing tau assemblies near the nucleus and examined cell viability. Nuclear transport was disturbed in this preparation. The combination of location, transport assays, and survival measurements addressed a possible route from altered tau organization to cellular dysfunction.

The selected original recording follows cells with nuclear-envelope and nearby tau inclusions as they undergo apoptotic death. Apoptosis is an organized process through which a cell dismantles itself. The event is measured in a cultured reporter system rather than inferred from the color of a tissue stain.

The experiment does not establish that every neuron with CTE tau dies through the same sequence. It identifies a testable cellular vulnerability involving protein organization and compartmental exchange. Human CTE tissue studies must determine which affected cell populations also exhibit the corresponding transport or survival changes.



Kang et al. (2021), Fig. 4.



Tau inclusions during cell death. Sang-Gyun Kang (2021).

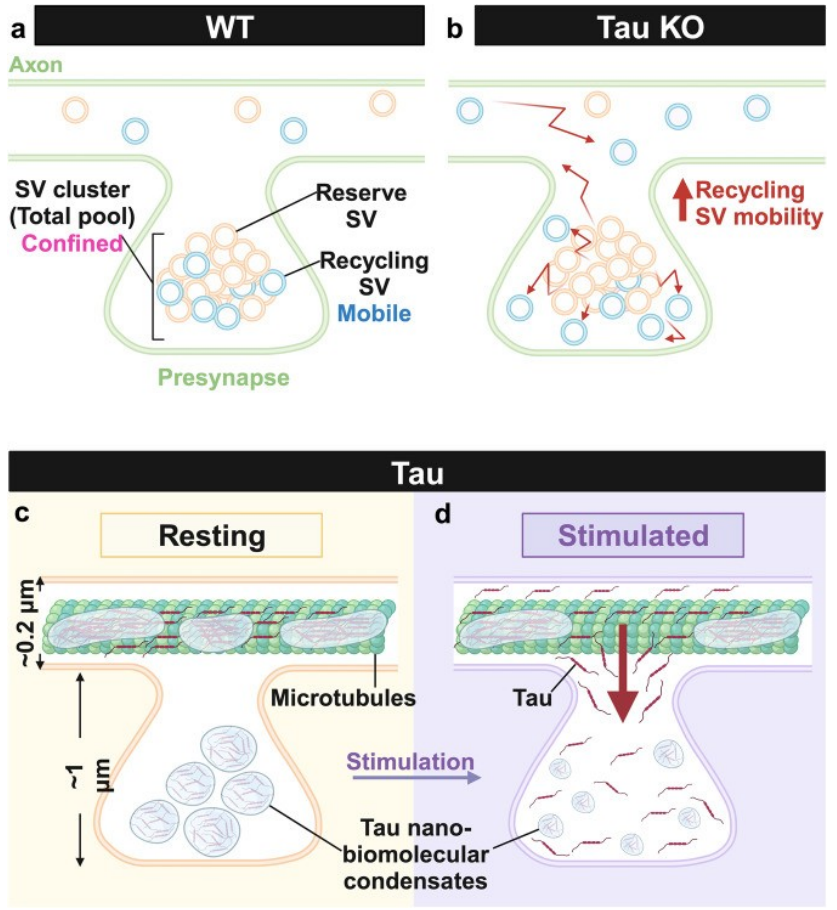
Play video

Synaptic vesicles make chemical communication possible

A synaptic vesicle is a small membrane-bound packet containing neurotransmitter. When an electrical signal reaches a presynaptic terminal, calcium entry helps trigger vesicle fusion with the cell membrane. Transmitter is released and can activate receptors on the receiving cell.

After release, membrane and vesicle proteins are retrieved and reused. Some vesicles readily participate in this recycling pool, while others are less likely to be released immediately. The organization and mobility of these populations influence how a terminal maintains communication across repeated activity.

Tau is present near some presynaptic structures as well as along axonal microtubules. Researchers therefore test whether changing tau alters the local organization of vesicles. A protein's normal synaptic role must be understood before its pathological accumulation can be interpreted as a simple loss of all useful function.



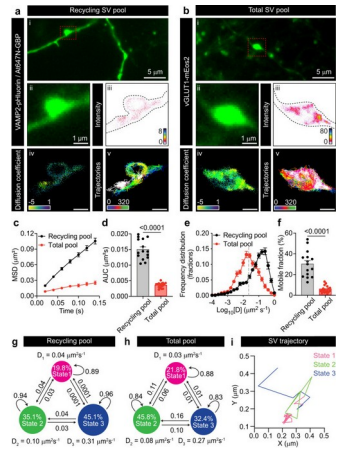
Longfield et al. (2023), Fig. 7.

Recycling vesicles form a regulated presynaptic pool

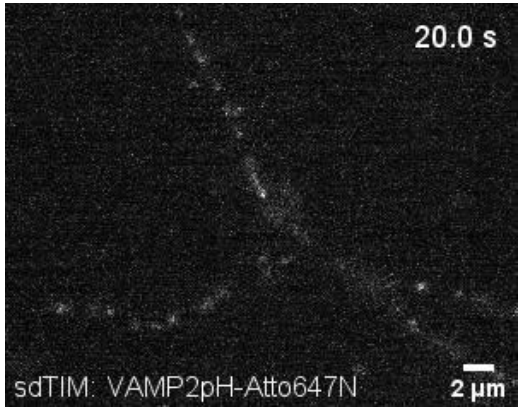
Researchers labeled vesicles that had participated in release and recycling in cultured neurons. A nanoscale imaging method followed their movement within presynaptic terminals. The label selected a functional subset of vesicles rather than assuming that every visible vesicle had the same release history.

The original recording follows the acquisition used to localize recycling-vesicle signals over time. Repeated observations allow trajectories and local clusters to be estimated. The changing fluorescence concerns vesicle-associated labels, not a measurement of a football player's memory or a direct image of CTE pathology.

The study identified activity-regulated tau nanoclusters that influenced this vesicle population. Tau's behavior within the terminal therefore provided a mechanism for changing chemical communication. A pathological tau alteration could affect signaling through such local organization as well as through failure of long-distance axonal transport.



Longfield et al. (2023), Fig. 1.



Recycling synaptic vesicle trajectories. Shanley F Longfield (2023).

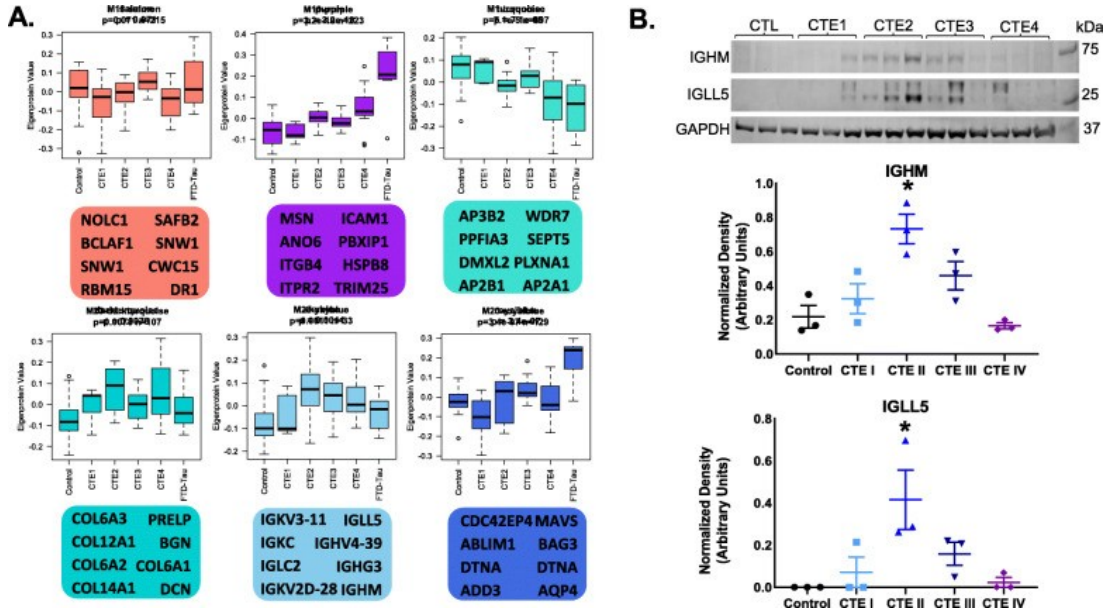
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CTE proteomics identifies several affected cell systems

A human postmortem study measured thousands of proteins across CTE pathological stages and compared them with control tissue and another tauopathy. Proteomics means examining the protein composition of a sample. The researchers grouped proteins that changed together into modules reflecting related biological functions.

CTE showed reductions in neuronal and synaptic protein modules alongside increases in inflammation-related and glial modules. Some changes overlapped those in other neurodegenerative diseases. The molecular phenotype therefore included altered communication machinery and support-cell responses rather than tau accumulation alone.

The study also identified CTE-associated modules enriched in immunoglobulins and extracellular-matrix proteins. The extracellular matrix is material surrounding cells that contributes to their physical and chemical environment. Changes in these proteins extended the observed pathology to the tissue context in which neurons and vessels operate.



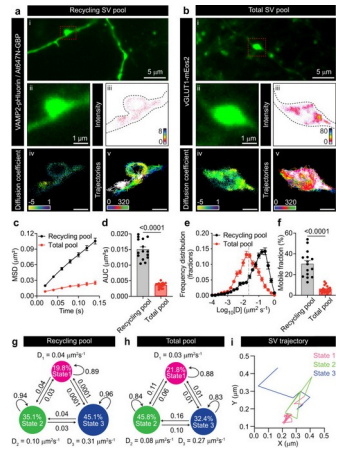
Gutierrez-Quiceno et al. (2021), Fig. 2.

Total and recycling vesicle pools move differently

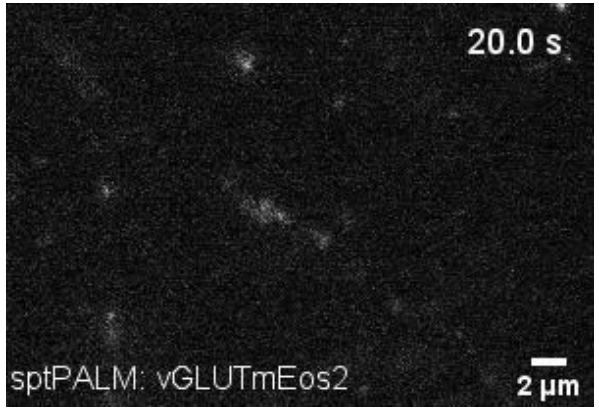
The synaptic tau study used a second labeling strategy to follow the broader population of glutamate-containing vesicles. A tagged vesicular transporter marked this pool independently of recent recycling. This allowed two populations within the same type of terminal to be compared with different molecular labels.

The original tracking movie follows the total pool rather than the recycling subset presented earlier. The populations displayed different mobility despite occupying a shared presynaptic cluster. A terminal is therefore not one uniform bag of vesicles moving and releasing with identical probabilities.

Tau condensates preferentially regulated the organization of recycling vesicles in the study. Comparing the two labeling strategies established this selectivity. The effect concerns a defined presynaptic process and does not require tau to control every vesicle population or every stage of neurotransmitter release equally.



Longfield et al. (2023), Fig. 1.



Total synaptic vesicle trajectories. Shanley F Longfield (2023).

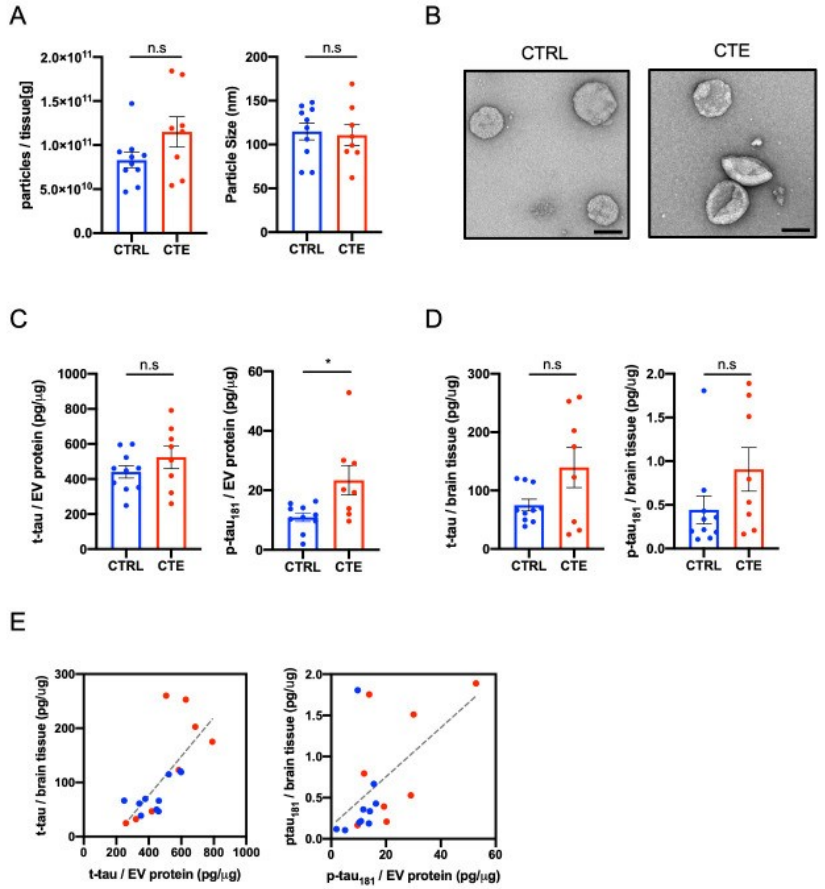
Play video

CTE brain-derived vesicles contain altered protein cargo

Researchers isolated extracellular vesicles from CTE and comparison brain tissue and measured their protein cargo. Extracellular vesicles are membrane-bound particles released into the space outside cells. They can carry proteins that report the state of their source tissue and may influence receiving cells.

CTE-derived vesicles contained more tau phosphorylated at threonine 181 in the tested samples. Tau-related protein modules also included molecules involved in synaptic secretion and membrane fusion. These findings identified altered cargo in brain-derived particles, rather than establishing a ready-to-use blood test.

The material came from postmortem brain tissue, which differs from a sample collected from a living player's circulation. Its composition provides molecular evidence about the affected brain environment. Testing whether the same information can be recovered reliably from blood or other fluids requires a separate clinical assay.



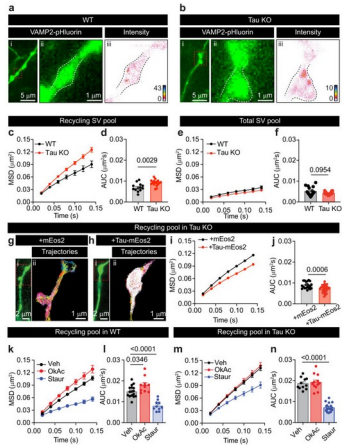
Muraoka et al. (2021), Fig. 1.

Tau and recycling-vesicle signals can be followed together

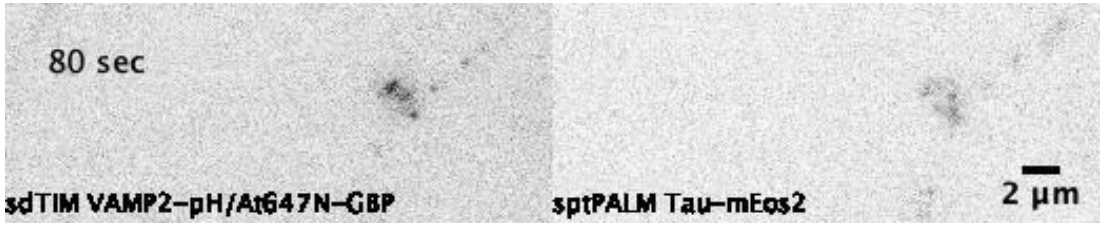
The investigators combined two imaging strategies to observe tau molecules and recycling vesicles in the same presynaptic preparation. Dual-color measurements distinguished the two labels while preserving their spatial relationship. This reduced the need to infer colocalization from experiments performed in unrelated cells.

The original recording follows tau-related and vesicle-related signals during the paired acquisition. Neuronal activity regulated tau diffusion and accumulation at presynapses in the study. Tau formed small condensates that altered the local mobility and organization of the recycling-vesicle population.

These results describe a normal activity-dependent function that can be compared with disease-associated tau changes. They do not equate a small physiological tau cluster with a CTE filament. The measured variables are molecular location and vesicle movement, providing concrete processes through which altered tau could influence synaptic output.



Longfield et al. (2023), Fig. 3.



Tau and recycling-vesicle trajectories. Shanley F Longfield (2023).

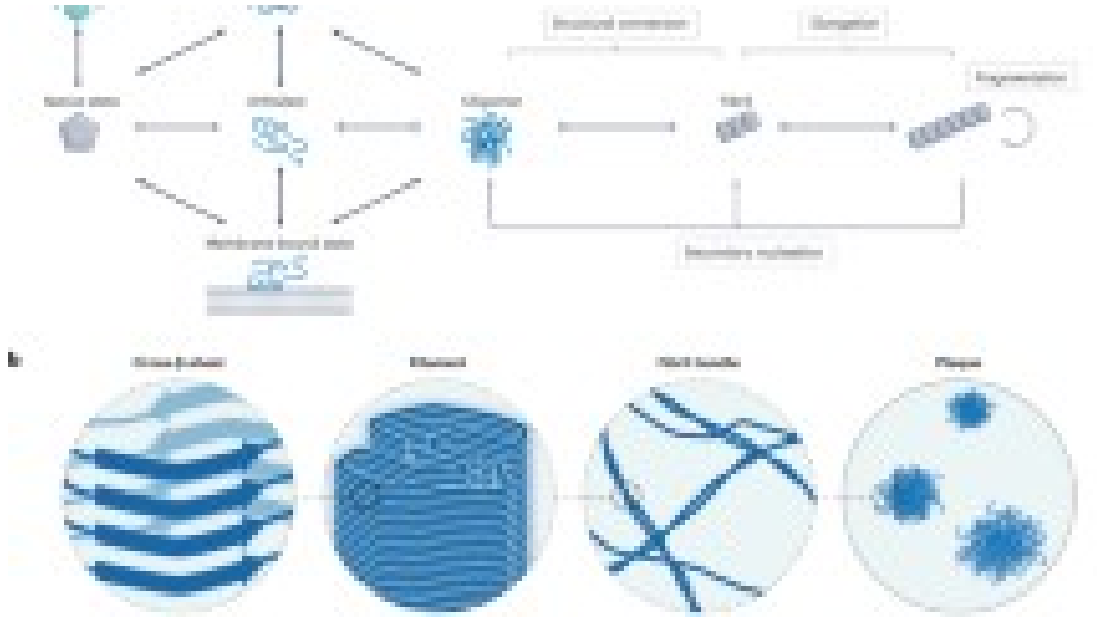
Play video

Condensation differs from fibril formation

Phase separation concentrates molecules into a compartment without requiring a surrounding membrane. A liquid-like condensate can exchange molecules with its surroundings and sometimes fuse with another droplet. This behavior can organize useful reactions as well as create conditions favorable to later aggregation.

A fibril instead contains molecules arranged in an extended, ordered assembly. Condensation and fibril formation need not occur simultaneously or depend on exactly the same interactions. A rounded droplet in a microscope is therefore not equivalent to a mature tau filament in CTE tissue.

Concentrating tau can increase opportunities for molecular encounters, but concentration alone does not specify the resulting fold. Sequence, modifications, cofactors, and the local environment influence assembly. Structural measurements are needed to identify whether an experimentally produced filament matches material extracted from a particular human disease.



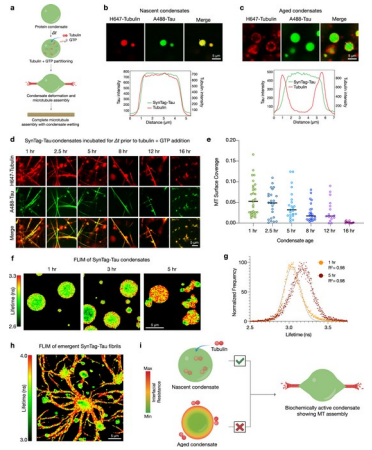
Mahendran et al. (2026), Fig. 1.

An engineered tau condensate can mature into fibrils

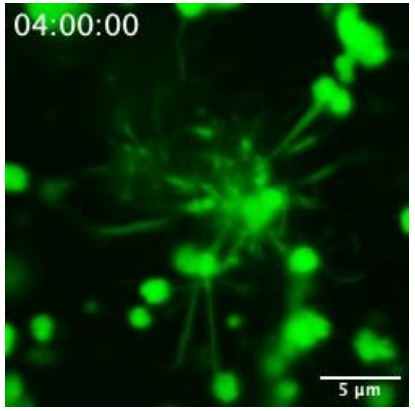
A recent study engineered tau to form condensates with measurable biochemical activity. The concentrated protein could recruit tubulin, the building block of microtubules, and promote microtubule assembly. Researchers then followed how these compartments changed as they aged under controlled conditions.

The original fluorescence recording tracks the time-dependent conversion of the engineered tau condensates into fibrillar structures. The study identified condensate interfaces as favorable sites for fibril nucleation. Nucleation is the formation of a small assembly that can support further growth.

The metabolite L-arginine delayed amyloid formation without simply abolishing phase separation in this system. That separation preserved condensate-related activity under the tested conditions. The experiment demonstrates a physical distinction that can guide mechanism research; it is not evidence that taking arginine treats CTE in people.



Mahendran et al. (2026), Fig. 2.



Engineered tau condensates and fibrils. Tharun Selvam Mahendran (2026).

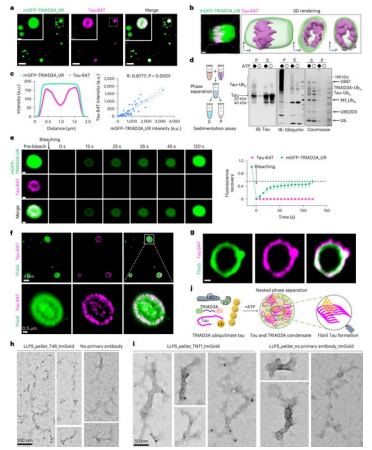
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An autophagy adaptor can concentrate tau for removal

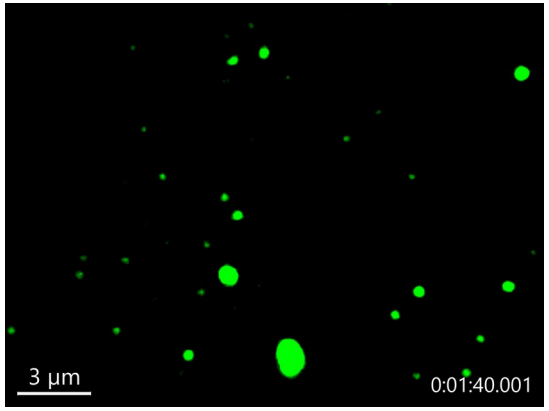
TRIAD3A is a protein that helps attach ubiquitin chains and direct cargo toward autophagy. Researchers found that it forms liquid-like droplets that capture tau. The original recording follows these adaptor droplets in a purified preparation, not a brain section from a player with CTE.

Tau inside the droplets could rapidly become fibrillar while also being directed toward degradation through autophagy-related interactions. Capturing cargo and converting its physical state therefore occurred together in the study. A structure associated with aggregation can participate in disposal rather than serving only as a permanent deposit.

In a tauopathy mouse model, ordinary TRIAD3A reduced phosphorylated-tau accumulation, whereas a disease-associated mutant worsened pathology and gliosis. Gliosis is a change in support-cell state during injury or disease. The findings identify a protein-handling mechanism whose consequences depend on successful processing of the captured cargo.



Jiechao et al. (2024), Fig. 6.



Purified TRIAD3A droplets. Zhou (2024).

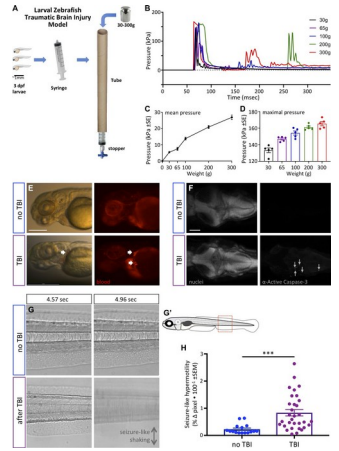
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Brain injury can disturb local blood flow

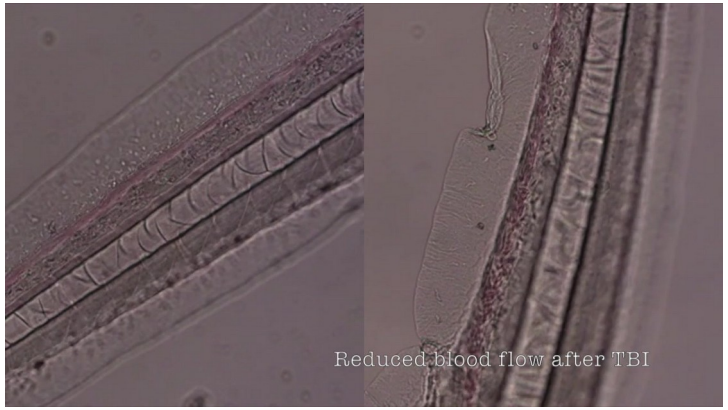
Researchers developed a traumatic brain injury model in larval zebrafish carrying a fluorescent tau reporter. The transparent preparation allowed vascular and cellular events to be observed after injury. This made it possible to examine a sequence connecting the physical insult with later tau-related changes.

The original recording follows an injury-associated blood-flow abnormality. Movement of blood cells differs from a static picture of the vessel wall, providing a direct observation of impaired circulation. The preparation tests acute injury biology rather than reproducing the full anatomy of a human cortical CTE lesion.

In the same model, injury was followed by cell death, abnormal neural activity, and reporter tau inclusions. These were measured as separate outcomes. The investigators could therefore test interventions against specific steps instead of assuming that altered blood flow, seizures, and protein aggregation were interchangeable signs.



Alyenbaawi et al. (2021), Fig. 3.



Zebrafish blood flow after brain injury. Hadeel Alyenbaawi (2021).

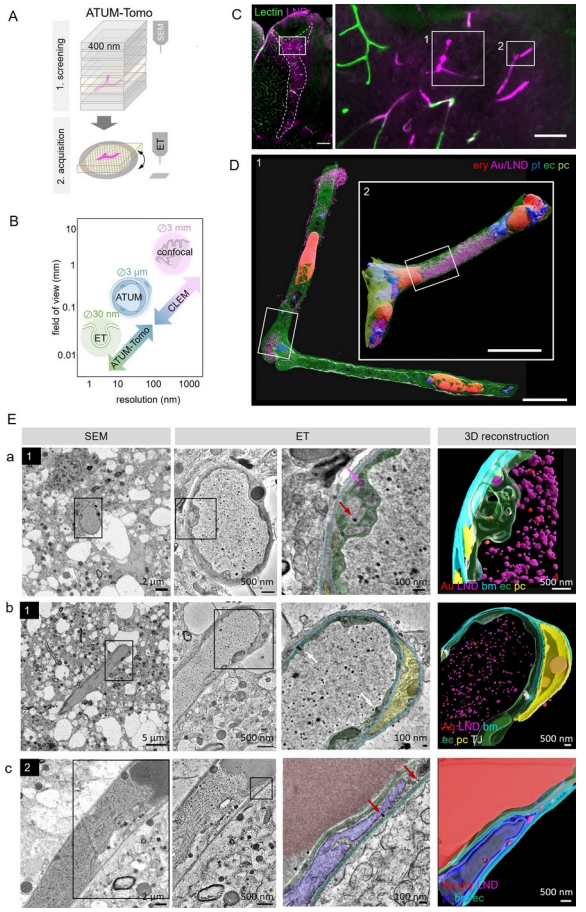
Play video

The blood-brain barrier is a cellular interface

The blood-brain barrier regulates exchange between circulating blood and brain tissue. Endothelial cells line the vessels and form specialized connections that restrict movement between neighboring cells. Pericytes, surrounding support cells, and astrocytic endfeet contribute to the conditions that maintain this interface.

A tight junction limits passage through the space between endothelial cells. Transporters and vesicular routes regulate other forms of movement across the lining. The barrier therefore performs selective exchange rather than acting as an inert wall that prevents every substance from reaching the brain.

Mechanical injury can disturb vessels, endothelial membranes, and interactions among the supporting cells. The consequences can include altered exposure to circulating molecules and impaired nutrient delivery. These events concern the environment surrounding neurons as well as direct physical damage to the neurons themselves.



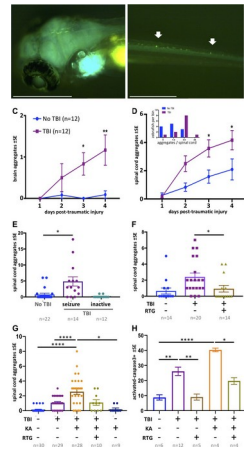
Kislinger et al. (2024), Fig. 3.

Abnormal activity contributes to injury-induced tau changes

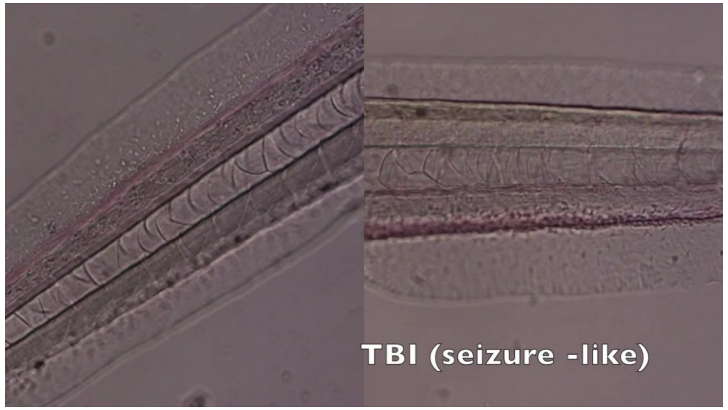
The zebrafish study examined post-traumatic seizures alongside blood-flow changes and tau-reporter inclusions. A seizure is an episode of abnormal synchronized neural activity. The original recording follows the injury-associated behavior and vascular disturbance in the transparent larval preparation.

Anticonvulsant treatments reduced tau-related pathology and cell death in the tested model. Blocking dynamin-dependent membrane processes also reduced these outcomes. Dynamin helps membrane compartments separate during endocytosis, a process that can influence how extracellular material enters a cell.

These manipulations tested potentially modifiable steps after trauma rather than preventing the original force. The results identify an interaction between neural activity, uptake-related processes, and tau accumulation in this preparation. They do not establish that every case of human CTE requires a recognized clinical seizure.



Alyenbaawi et al. (2021), Fig. 5.



Zebrafish seizure-like behavior after injury. Hadeel Alyenbaawi (2021).

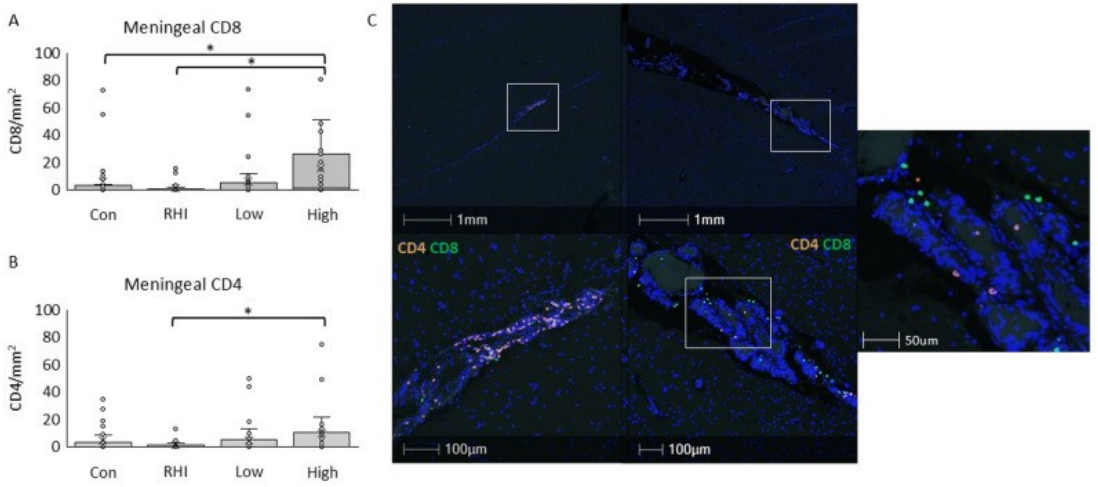
Play video

Immune responses involve cells with different jobs

Microglia are resident immune cells within brain tissue. Macrophages also occupy the meninges and regions around vessels, while lymphocytes can enter or reside in associated compartments. These populations share some immune functions but are not interchangeable names for one cell type.

T cells use specialized receptors to recognize molecular targets presented by other cells. Some T cells coordinate responses, while others can attack targeted cells. Their presence near a vessel or neuron identifies an immune population, but does not by itself reveal which target the cells recognized.

A chronic tissue response can include resident-cell changes and recruitment from outside the local compartment. Researchers therefore identify cell markers and anatomical location together. Distinguishing meningeal, perivascular, and parenchymal cells helps determine where immune activity accompanies CTE-related pathology.



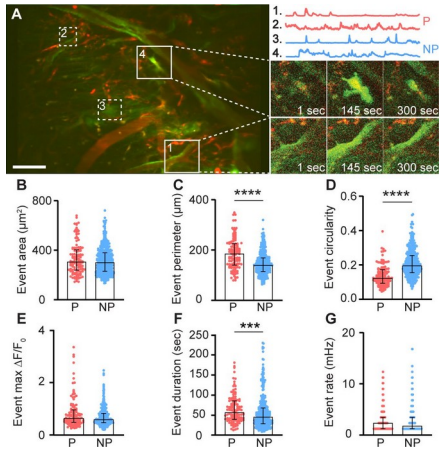
Calderazzo et al. (2026), Fig. 1.

Meningeal calcium signals depend on cell location

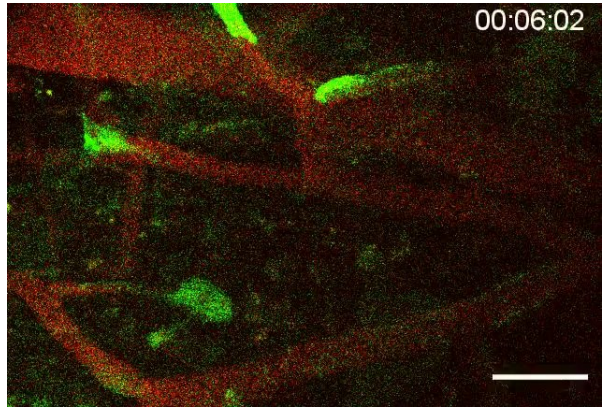
A recent mouse study imaged macrophages in the meninges of awake animals. The cells expressed a fluorescent calcium indicator, allowing intracellular activity to be followed over time. Researchers distinguished macrophages beside blood vessels from those occupying other parts of the same tissue compartment.

The original recording follows spontaneous calcium events in both locations. The groups differed in features including event duration and the timing of their signals. Calcium fluorescence therefore supplied a dynamic measurement of cell state beyond simply counting how many macrophages were present.

Some vessel-associated macrophages showed activity coupled to changes in vessel diameter during movement. The finding identifies communication between the local vascular environment and immune-cell activity. This is a physiological meningeal preparation, providing a comparison for the perivascular abnormalities examined in human CTE tissue.



Carneiro-Nascimento et al. (2026), Fig. 2.



Mouse meningeal macrophage calcium. Simone Carneiro-Nascimento (2026).

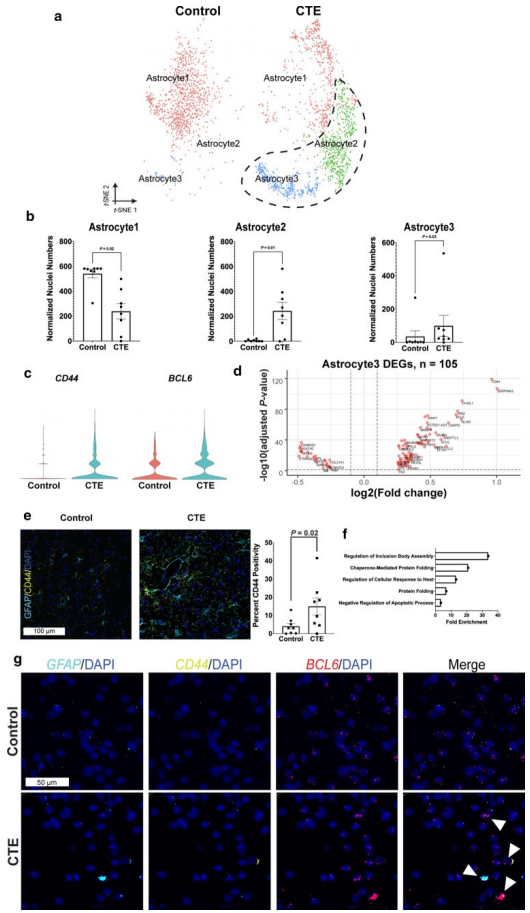
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Glia can protect tissue and contribute to its dysfunction

Glial cells support neurons through several distinct processes. Astrocytes regulate extracellular conditions and metabolic support, oligodendrocytes maintain myelin, and microglia survey tissue and remove selected material. A change in one of these populations can affect communication without beginning inside the affected neuron.

Reactivity describes changes in a cell's molecular state, shape, or function during a challenge. Reactive cells can help contain damage, but particular responses can also promote inflammation or inappropriate removal of useful structures. A single marker does not classify every reactive glial cell as protective or harmful.

CTE tissue can therefore contain multiple interacting abnormalities around the tau lesion. Measuring the cell populations separately helps identify which processes changed and where they occurred. The biological effect depends on the response being carried out, not merely whether a cell has become larger or expresses more of one protein.



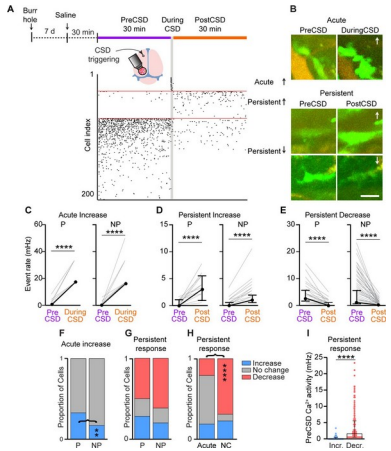
Chancellor et al. (2021), Fig. 4.

Cortical disturbance changes meningeal cell activity

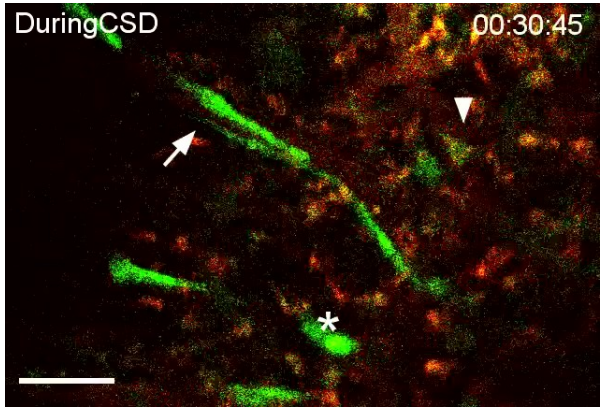
Cortical spreading depolarization is a wave of intense disturbance in neuronal and glial membrane potentials. Researchers induced this event while imaging meningeal macrophages. The experiment tested whether activity in brain tissue altered the behavior of immune cells in the surrounding meningeal compartment.

The selected original recording follows calcium activity before and after the disturbance. Most measured macrophages showed a persistent decrease, while a smaller subset showed increased signals. The response was therefore heterogeneous rather than every nearby immune cell becoming uniformly more active.

Calcitonin gene-related peptide receptor signaling contributed to the increased responses but did not explain the widespread decrease. A signal can regulate one branch of the response without controlling all of it. The preparation identifies a pathway for brain activity to alter a vascular-associated immune environment.



Carneiro-Nascimento et al. (2026), Fig. 6.



Meningeal calcium before and after cortical disturbance. Simone Carneiro-Nascimento (2026).

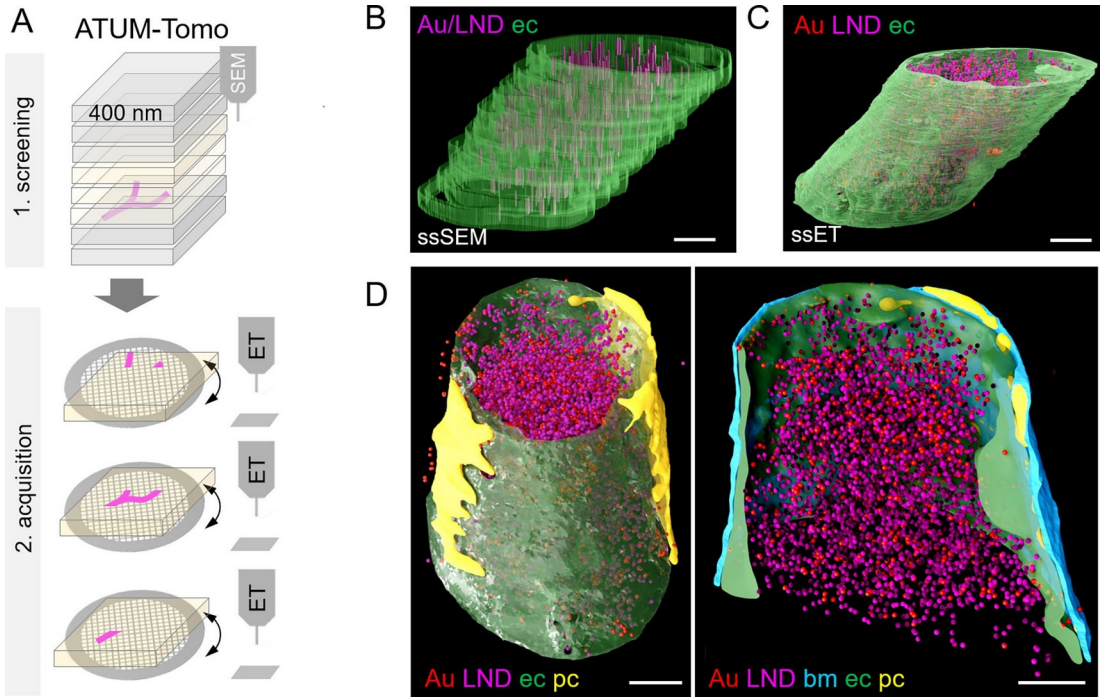
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Perivascular spaces organize exchange around vessels

Perivascular spaces are fluid-containing compartments associated with vessels entering or traversing brain tissue. Their organization depends on vessel walls and surrounding cellular structures. Fluid and solute movement in these regions contributes to exchanges between the brain, its surfaces, and the circulation.

Astrocytic endfeet surround many vessels and help organize this boundary. Endfeet contain proteins involved in water movement and local homeostasis. Changes in their arrangement could alter the environment in which neurons and immune cells operate, but tissue structure and fluid transport must be measured separately.

The perivascular location of a CTE lesion makes these compartments relevant to its neurobiology. It does not establish that one drainage defect explains every lesion. Human imaging can measure enlarged spaces, while microscopic studies and tracer experiments test different components of exchange around the vessel.



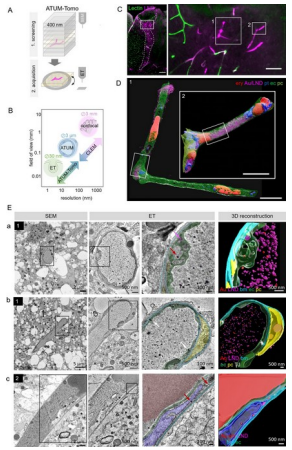
Kislinger et al. (2024), Fig. 4.

Electron tomography resolves vascular injury

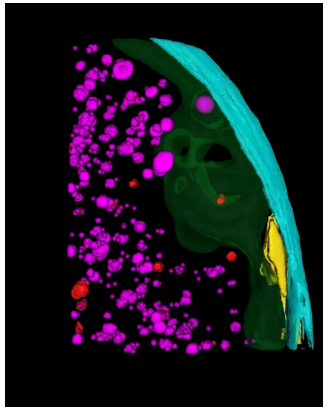
Researchers combined fluorescence imaging with electron microscopy in a mouse brain-injury model. Fluorescent and electron-dense particles marked regions associated with vascular disturbance. The investigators then found the same locations at higher resolution to examine endothelial cells and surrounding structures.

The original published reconstruction rotates a selected vascular region containing particles within an endothelial swelling. The colors identify segmented structures from the microscopy data. It is an anatomical reconstruction, not real-time footage of a particle traveling through the blood-brain barrier.

The broader analysis also identified preserved junctions at some locations and disrupted endothelium at others. A barrier abnormality can therefore involve more than one structural route. Resolving the lining, junctions, and internal compartments helps distinguish those possibilities within an injured vessel.



Kislinger et al. (2024), Fig. 3.



Electron-microscopy vascular reconstruction. Georg Kislinger (2024).

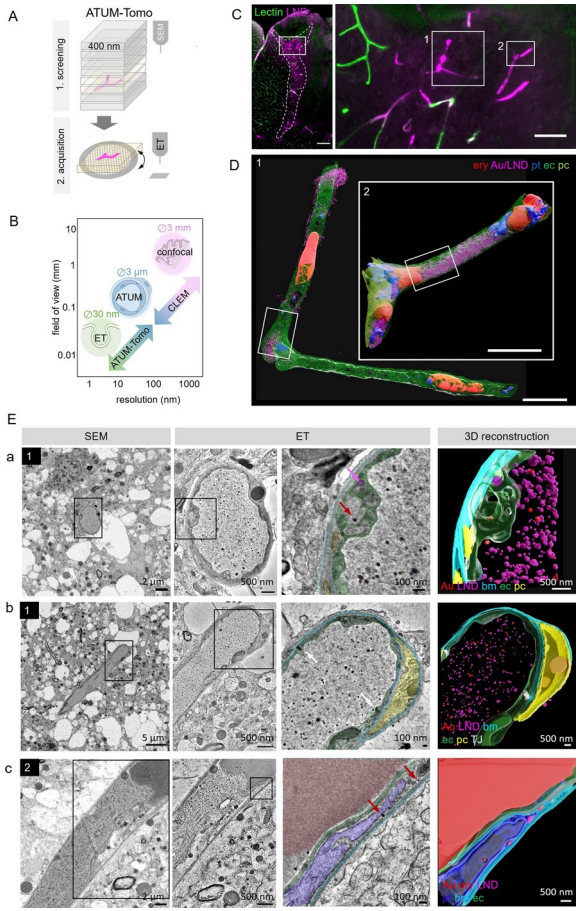
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Vesicles provide a route through the endothelial cell

Transcytosis carries material across a cell through membrane-bound compartments. Endocytosis first brings cargo into the cell, internal trafficking moves it between compartments, and release can occur at another membrane surface. This route differs from movement through a gap between neighboring endothelial cells.

A vessel can therefore change its exchange properties without every tight junction visibly opening. The type and destination of internalized cargo influence whether material reaches the surrounding tissue. Finding particles inside an endothelial cell identifies uptake, while demonstrating passage across the entire barrier requires additional evidence.

Ultrastructural imaging can locate vesicles and membranes at very high resolution but captures tissue at a particular time. Tracers and physiological measurements add information about movement and rate. Combining the approaches distinguishes a structural opportunity for transport from a directly measured flux across the vessel wall.



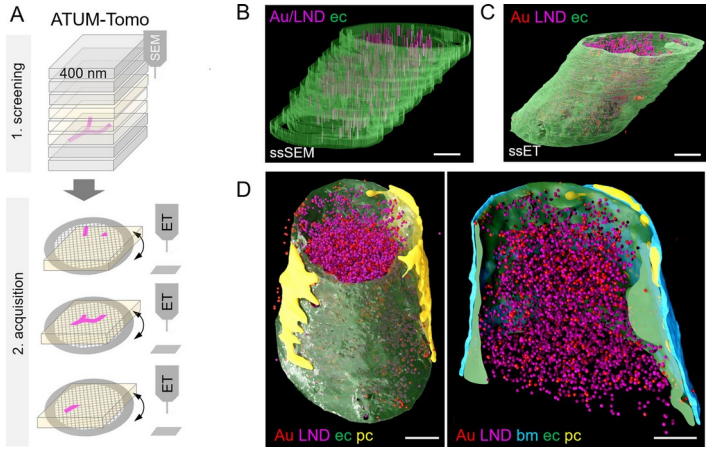
Kislinger et al. (2024), Fig. 3.

Serial images reconstruct the vascular neighborhood

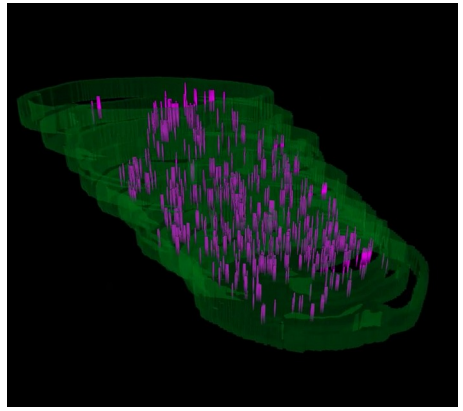
The vascular study developed a method that allowed selected tissue sections to be transferred from a screening preparation into electron tomography. Screening identified a rare region of interest, while tomography resolved much finer structures. The method combined tissue context with detailed examination of local membranes.

The selected original reconstruction follows a vascular region across consecutive sections. Endothelium, a pericyte, a platelet, and the basement membrane occupy different positions within the reconstructed volume. The basement membrane is a supporting extracellular layer surrounding the vascular cells.

A three-dimensional reconstruction reduces the risk of mistaking a structure's appearance in one thin section for its complete shape. The experiment makes the vascular neighborhood anatomically interpretable. The rotation displays recorded and segmented tissue, not an animation of blood flow or a simulation of tau spreading.



Kislinger et al. (2024), Fig. 4.



Serial-section vascular reconstruction. Georg Kislinger (2024).

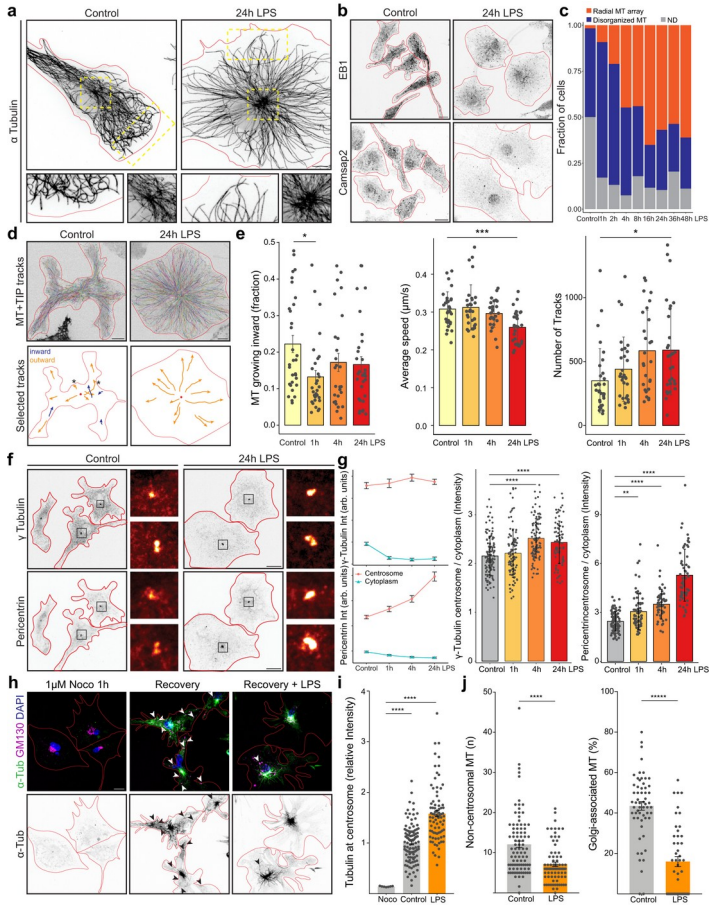
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Microglial reactivity requires structural remodeling

Microglia extend and retract processes while monitoring their surroundings. Changing shape requires reorganization of the cytoskeleton, including microtubules and actin. These structures support movement, intracellular delivery, and the formation of compartments used to collect material from outside the cell.

Inflammatory signals can alter how these components grow and where they are arranged. A microglial cell can become more mobile or change the direction in which it releases material. Its visible shape therefore reflects an active organization of structural proteins rather than a permanent category fixed at development.

In CTE, altered immune-cell states occur alongside neuronal and vascular abnormalities. Mechanistic experiments can test how a particular structural change alters secretion or movement. The molecular processes must be measured before a rounded or branched appearance is assigned one universal effect on nearby neurons.



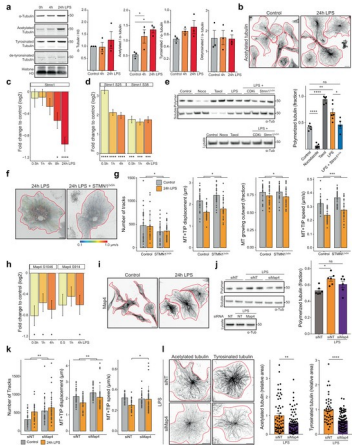
Max et al. (2023), Fig. 4.

Inflammatory stimulation polarizes microglial microtubules

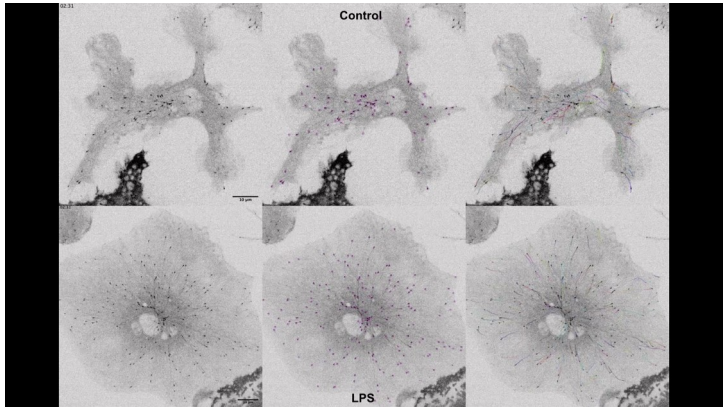
Researchers imaged the growing ends of microtubules in primary microglial cells. A fluorescent marker followed the growing tips, allowing the direction and organization of microtubule growth to be measured. Cells exposed to lipopolysaccharide were compared with untreated controls.

The original recording contains raw images, detected growing tips, and their computed tracks. Lipopolysaccharide is a bacterial component used here as a controlled inflammatory stimulus, not evidence that CTE is an infection. The experiment isolates a response to inflammatory signaling in a cellular preparation.

Reactive cells reorganized microtubules into a more polarized arrangement associated with altered morphology and cytokine release. Cytokines are proteins that influence the activity of other cells. The structural changes therefore affected both the cell's organization and the way it delivered inflammatory signals to its surroundings.



Max et al. (2023), Fig. 5.



Microglial microtubules: control and LPS. Adrian (2023).

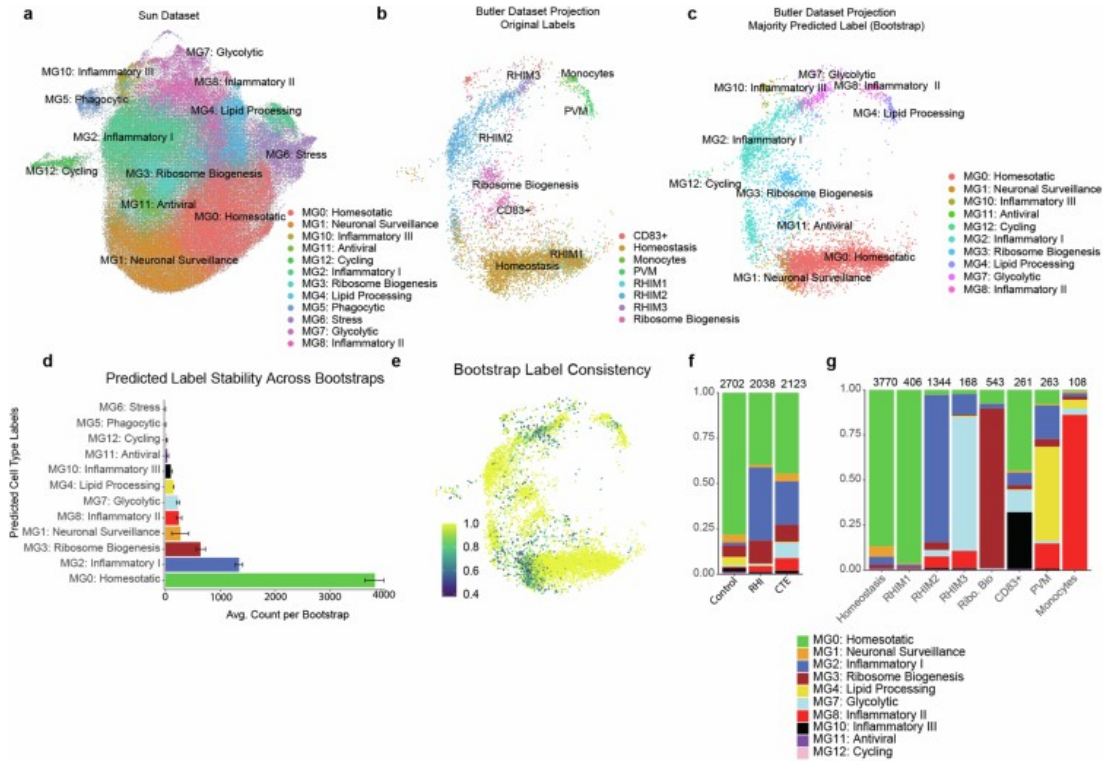
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Repeated impacts alter several cortical cell populations

A human study compared cortical tissue from unexposed controls, people with repeated head impacts without diagnostic CTE tau, and people with low-stage CTE. Single-nucleus RNA sequencing measured which genes individual nuclei were transcribing. This separated the responses of neurons, microglia, astrocytes, and vascular cells.

Repeated-impact groups contained inflammatory SPP1-expressing microglia, altered endothelial cells, and astrocytic changes. Some abnormalities were present even without the defining tau lesion. Their presence in that group identifies an exposure-associated cellular state rather than proving that each donor would later have developed CTE.

The study also found loss of superficial-layer neurons near sulcal depths and inferred altered signaling between microglia and endothelial cells, including a possible TGFβ1 pathway. These results identify a multicellular environment in which tau pathology can arise. The proposed communication signal remains distinct from a directly demonstrated causal treatment target.



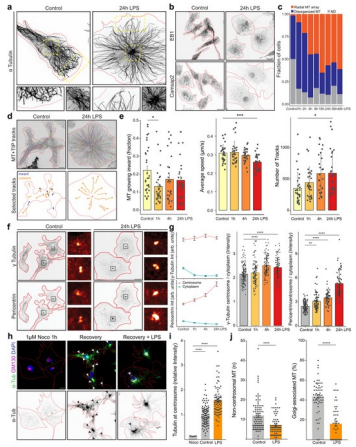
Butler et al. (2025), Fig. 5.

A washout experiment tests microtubule regrowth

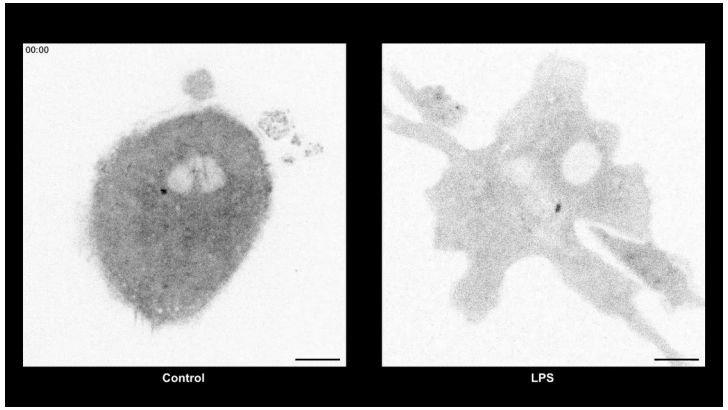
Nocodazole disrupts microtubule polymerization, reducing the formation of long microtubule structures. Removing the drug allows growth to resume. The microglial study used this washout approach to examine how prior inflammatory stimulation changed the organization of newly forming microtubules.

The original recording compares untreated and inflammation-primed cells after nocodazole removal. Growing-tip signals reveal where regrowth begins and how it is organized. This is a different measurement from simply comparing the steady-state shape of two cells before and after inflammatory stimulation.

The experiment helped identify altered microtubule nucleation and organization as part of the reactive state. Nucleation forms a starting site from which a microtubule can grow. Changing the placement of those starting sites can redirect intracellular delivery and contribute to the cell's polarized response.



Max et al. (2023), Fig. 4.



Microtubule regrowth after washout. Adrian (2023).

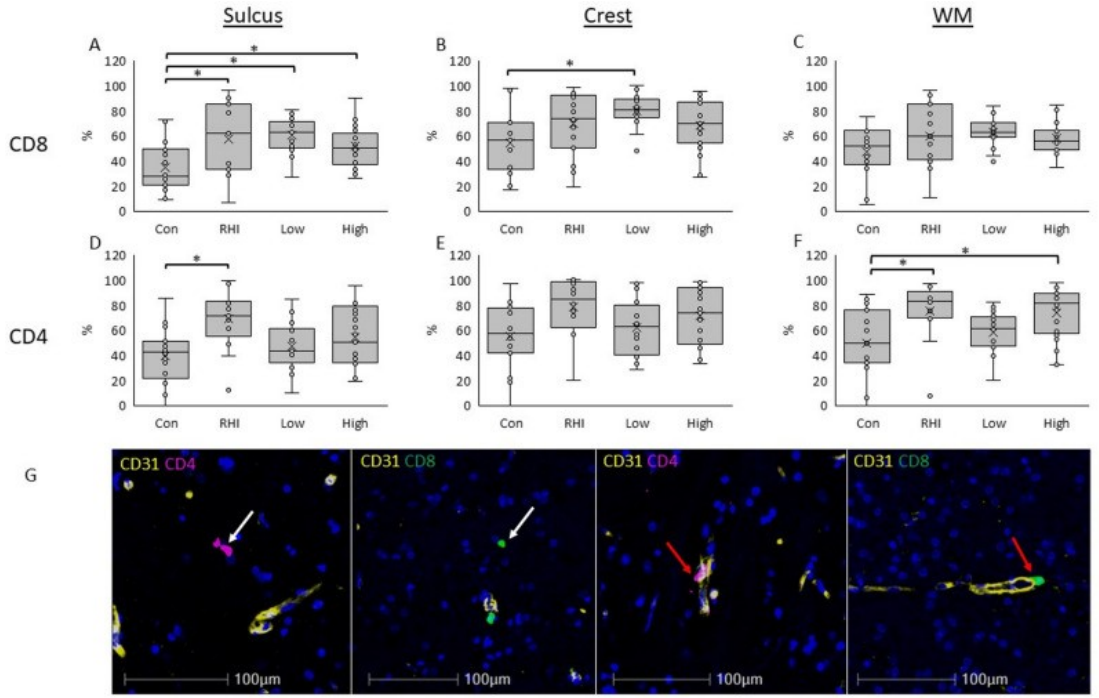
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Human CTE tissue contains altered T-cell neighborhoods

A human postmortem study identified T cells together with microglia, phosphorylated tau, and synaptic markers. Investigators compared the meninges, sulcal gray matter, gyral crest, and white matter. The design located immune cells within their anatomical setting rather than combining all brain compartments into one count.

T-cell infiltration was increased in sulcal tissue from repeated-impact groups, including donors without diagnostic CTE, and was associated with exposure duration and synaptic loss. Cell subsets differed across the pathological groups. These findings extended the inflammatory phenotype beyond resident microglia alone.

T cells also occupied neighborhoods containing tau deposits and cells expressing antigen-presentation machinery. This spatial association identifies possible interactions to test. It does not establish which molecular target the T cells recognized or that their presence alone caused the nearby neuronal changes.



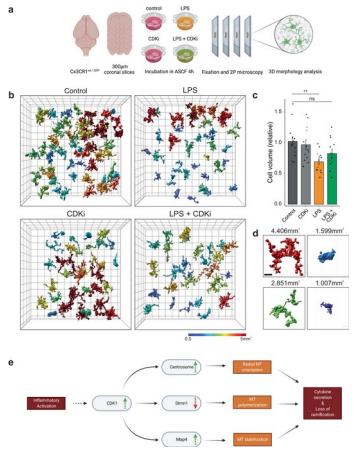
Calderazzo et al. (2026), Fig. 2.

Structural responses extend into intact brain tissue

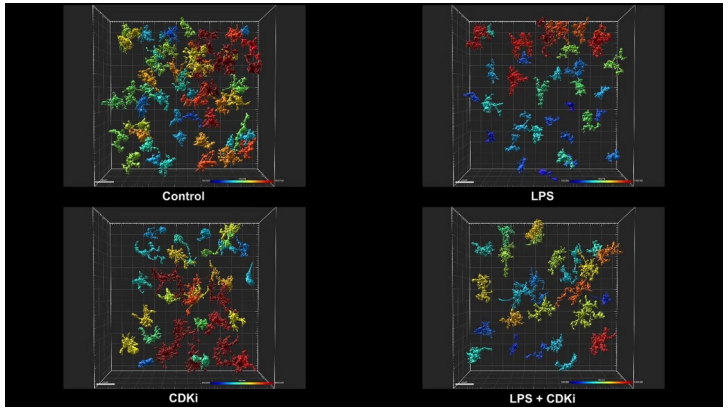
The microtubule-remodeling study also examined microglia in acute brain slices. Researchers compared control tissue, inflammatory stimulation, enzyme-inhibitor treatment, and the combined condition. This tested whether the cell-shape effects observed in culture were also present within a tissue environment.

The original three-dimensional image stacks preserve those four source conditions. Their changing viewpoint reveals cell shape and volume within the recorded tissue. The display is a reconstruction of stained biological structures, not a live recording of a human CTE lesion developing.

Inhibiting cyclin-dependent kinase activity altered the inflammatory structural and secretory response in the study. These enzymes regulate proteins through phosphorylation. The experiment therefore tested a controllable intracellular step, while human CTE work establishes whether comparable molecular states occur in the affected immune-cell populations.



Max et al. (2023), Fig. 7.



Microglial morphology in brain tissue. Adrian (2023).

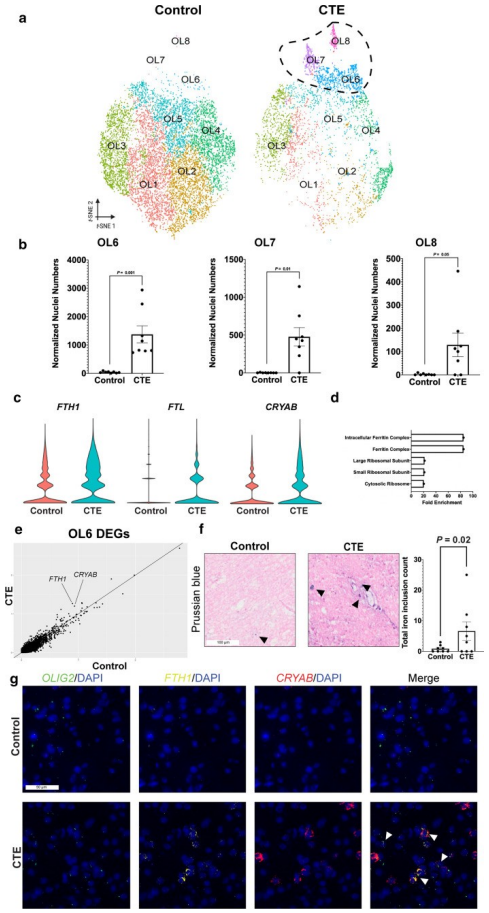
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White-matter glia show cell-specific changes in CTE

Researchers examined nuclei from human frontal white matter in CTE and comparison brains. White matter contains many myelinated axons, so its function depends strongly on oligodendrocytes and associated support cells. Sequencing each nucleus separated changes in these populations from the average signal of the whole sample.

CTE samples contained fewer oligodendrocytes and a changed balance among their subtypes. A CTE-enriched population expressed more genes related to iron handling and cellular stress. Independent staining also demonstrated increased iron accumulation in the tissue, providing a second kind of measurement.

Astrocyte numbers were not altered in the same way, but their gene expression showed increased inflammatory programs. Different glial populations therefore changed through different processes. A loss of myelin-supporting cells and a change in astrocytic state need not be represented by the same marker or one uniform gliosis score.



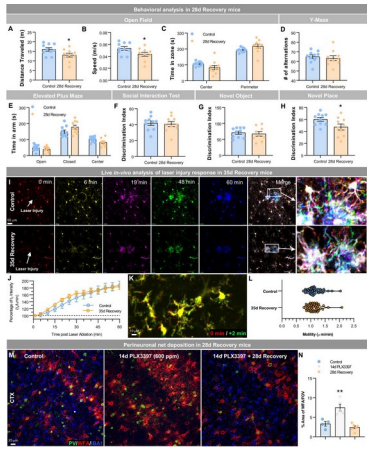
Chancellor et al. (2021), Fig. 3.

Microglia rapidly redirect processes toward focal damage

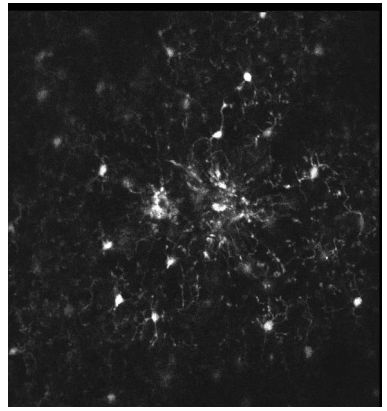
A mouse study of microglial depletion and recovery first characterized how ordinary cells respond to a small laser lesion. Live imaging followed their processes around the damaged site. This supplied a functional comparison for cells that later repopulated a depleted brain.

The original control recording follows processes extending toward the focal injury. Directed process movement is distinct from the entire cell body migrating to a new location. Both can change a cell's contact with surrounding structures, but they use different patterns of cytoskeletal reorganization.

An acute lesion isolates a local response over minutes rather than the long exposure history of CTE. It nevertheless provides an observable test of tissue surveillance and injury detection. Researchers can use such a defined challenge to compare whether experimentally altered immune cells retain those functions.



Hohsfield et al. (2021), Fig. 9.



Normal mouse microglial response to laser injury. Lindsay A Hohsfield (2021).

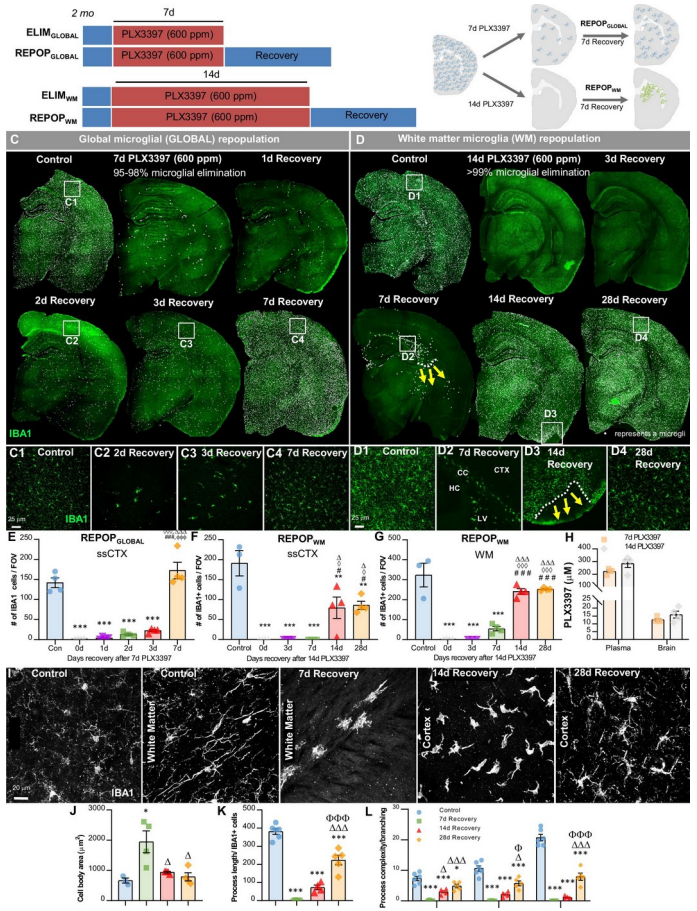
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Removing a cell population does not specify its replacement

Microglial survival depends on signals that include colony-stimulating factor 1 receptor activity. Blocking this receptor can deplete many microglia in experimental animals. The resulting tissue is not simply a normal brain with a permanently switched-off inflammatory marker.

Cells that survive or enter the emptied region can proliferate and repopulate it. Their origin, molecular state, and behavior may differ from the original population. An intervention that reduces cell number therefore requires follow-up measurements of which cells return and what functions they perform.

This distinction matters when considering immune targets for CTE neurobiology. Suppressing a harmful response is different from removing all cells capable of that response. Microglia also support normal tissue maintenance, so a proposed intervention must specify the cellular process it intends to change and the functions it preserves.



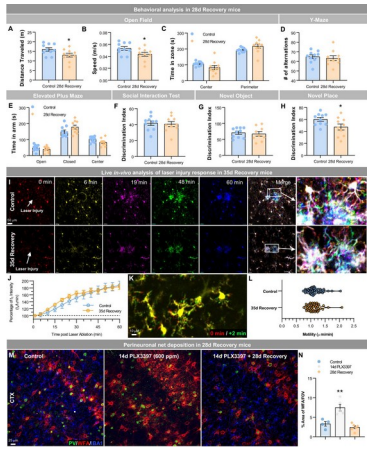
Hohsfield et al. (2021), Fig. 1.

Repopulating immune cells can retain an injury response

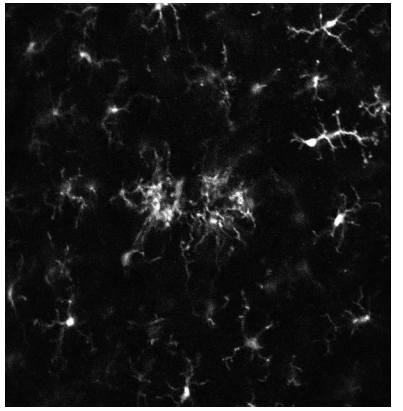
After prolonged receptor inhibition depleted microglia, the mouse study followed cells that repopulated the brain. Many arose from surviving cells associated with the subventricular region and white matter. Their spread formed a dynamic wave rather than every empty location being filled simultaneously.

The original recovery recording follows the repopulated cells responding to a focal laser lesion. Comparing it with the control tests a concrete surveillance function after replacement. The study found relatively few differences in several functional outcomes despite persistent molecular differences between the populations.

Cell number, gene expression, and functional behavior therefore required separate measurements. Restoring one did not establish that every property had returned to its original state. This experimental comparison defines what must be tested when immune-cell manipulation is proposed for chronic neurodegeneration, including CTE.



Hohsfield et al. (2021), Fig. 9.



Repopulated microglial response to laser injury. Lindsay A Hohsfield (2021).

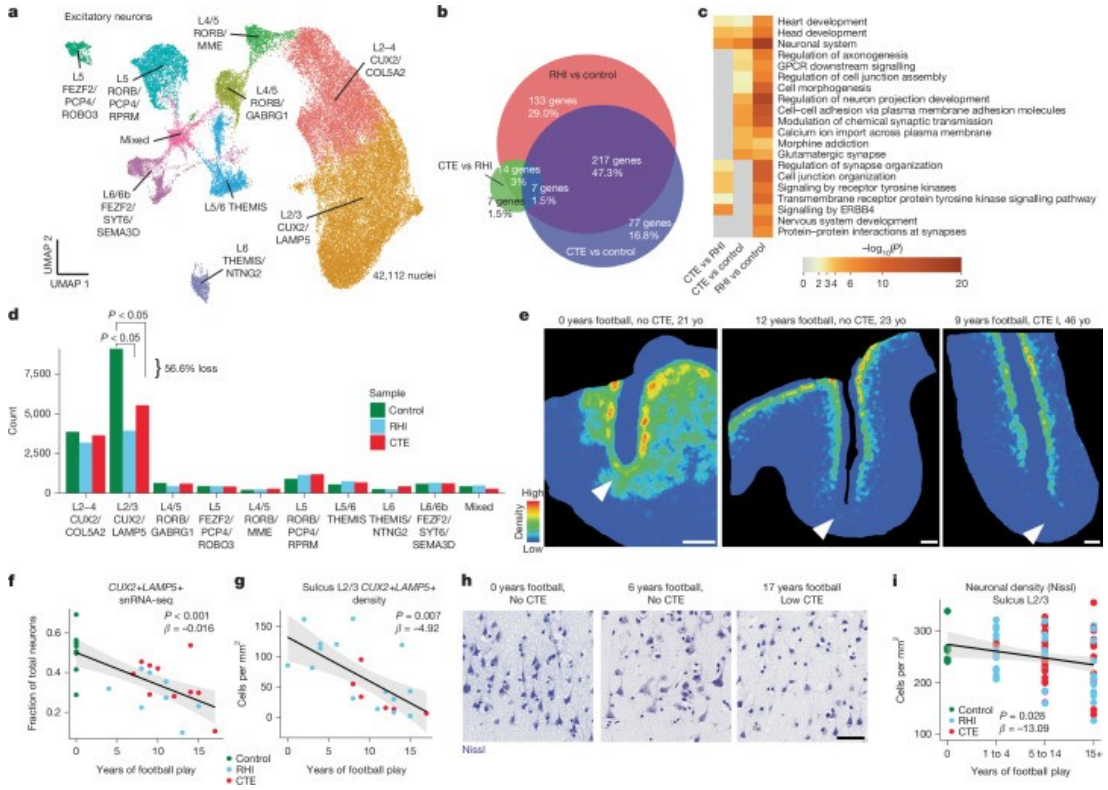
Play video

A symptom reflects activity across a neural circuit

Memory, planning, and emotional regulation depend on communication among connected neuronal populations. Damage to a projection or loss of synapses can change how information moves through that circuit. The resulting symptom is not determined solely by whether one neuron contains a visible tau deposit.

Cortical layer identity influences which inputs a neuron receives and where it sends signals. Losing a superficial-layer population can therefore affect a different set of interactions than losing cells deep in the same cortical column. Cell counts become more informative when combined with anatomical and molecular identity.

A player's reported cognitive difficulty must also be assessed alongside sleep, pain, mood, medication effects, and other neurological conditions. Those influences can change circuit function without reproducing the defining CTE lesion. Clinical assessment and neuropathological diagnosis answer related but different questions about the person's brain.



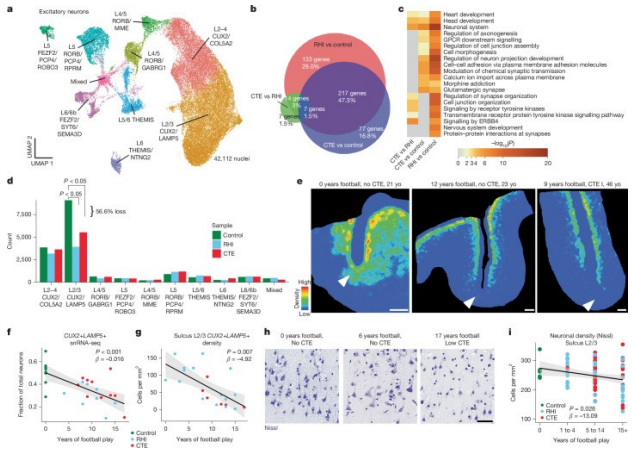
Butler et al. (2025), Fig. 4.

CTE studies distinguish neuronal loss from its clearance

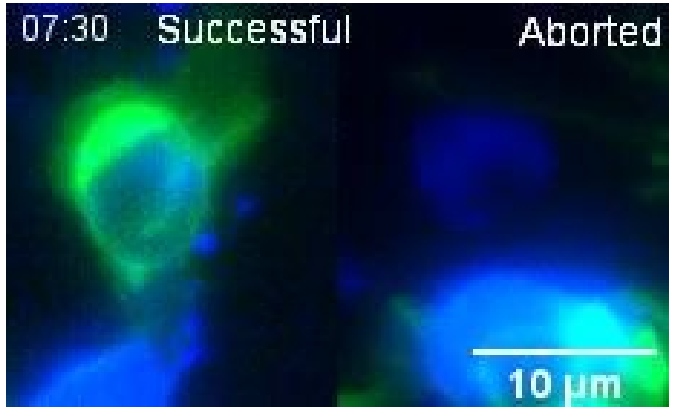
The young-donor cortical study identified reduced superficial-layer excitatory neurons near sulcal depths in repeated-impact groups. Researchers combined single-nucleus data with independent tissue measurements. The neuronal loss was not explained simply by restricting analysis to donors with recognizable diagnostic tau lesions.

The affected populations expressed markers including CUX2 and LAMP5, helping identify which cells were reduced. These markers specify a cellular population rather than proving why those cells were lost. The findings establish selective vulnerability that can be examined alongside vascular and inflammatory changes.

A separate live zebrafish preparation records successful and aborted attempts by microglia to engulf dying neurons. Engulfment is the physical collection of material into a cell. The filmed actions occur during clearance after neuronal death; it is not footage demonstrating that microglia killed the neurons missing from human CTE-related tissue.



Butler et al. (2025), Fig. 4.



Successful and aborted microglial engulfment. Katrin Möller (2022).

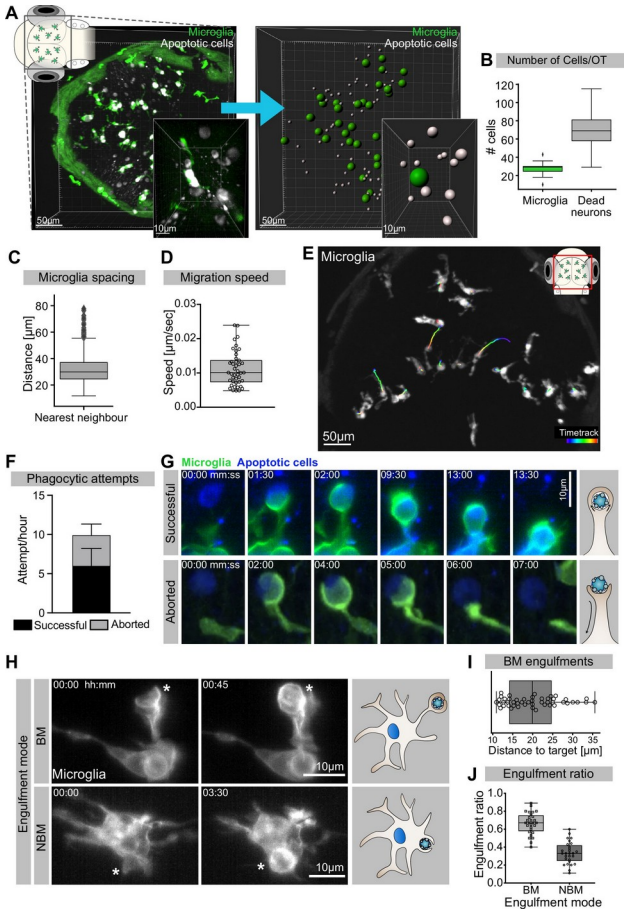
Play video

A centrosome helps organize intracellular delivery

The centrosome organizes many microtubules within a cell. Positioning this organizing center can influence the directions along which vesicles and other cargo move. A branched microglial cell must coordinate that internal delivery system with the particular process carrying out a response.

A phagosome forms when a cell encloses material it has engulfed. Building this compartment requires membrane movement and local organization, not just contact with the target. The cell must direct resources into one region while continuing other activities elsewhere in its body and branches.

This organization provides a cellular mechanism through which immune-cell shape influences function. A marker of a reactive state does not reveal which branch receives the required cargo. Live imaging can therefore test a process that static human tissue studies cannot reconstruct from one stained section alone.



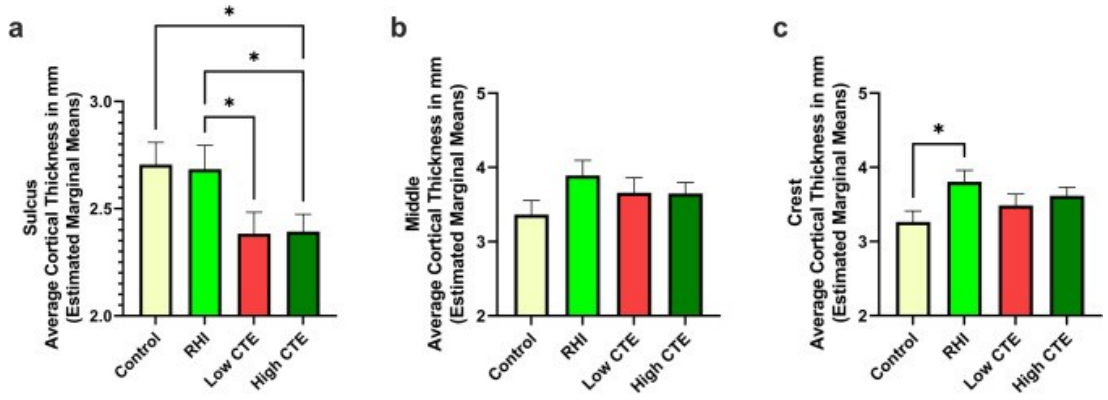
Möller et al. (2022), Fig. 1.

Sulcal neuronal loss differs from tissue thinning

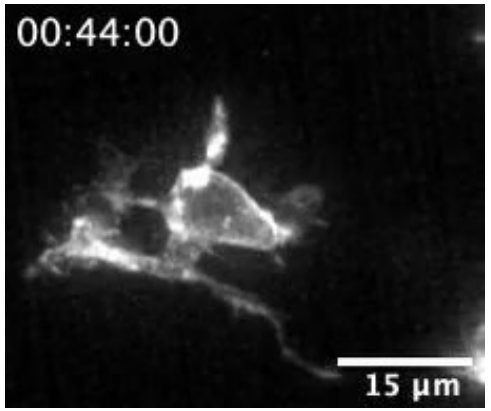
The regional human CTE study quantified neuronal density and cortical thickness separately. Both were reduced preferentially in sulcal tissue, but their relationships with exposure and tau were not identical. A thinner cortical region cannot therefore be interpreted automatically as a direct count of lost neurons.

Tissue thickness also depends on processes extending between cells and on the surrounding cellular and extracellular environment. The study's analyses suggested tau-related and partly tau-independent contributions to the degenerative measures. Separating the outcomes preserved information about those different routes to altered tissue structure.

The accompanying zebrafish recording follows a microglial cell clearing dead neurons sequentially. It makes the physical removal of cellular material visible in an independent model. This is a process that follows cell death, distinct from the measurements used to establish the amount and regional pattern of neuronal loss in CTE.



Nicks et al. (2024), Fig. 2.



Sequential clearance of dead neurons. Katrin Möller (2022).

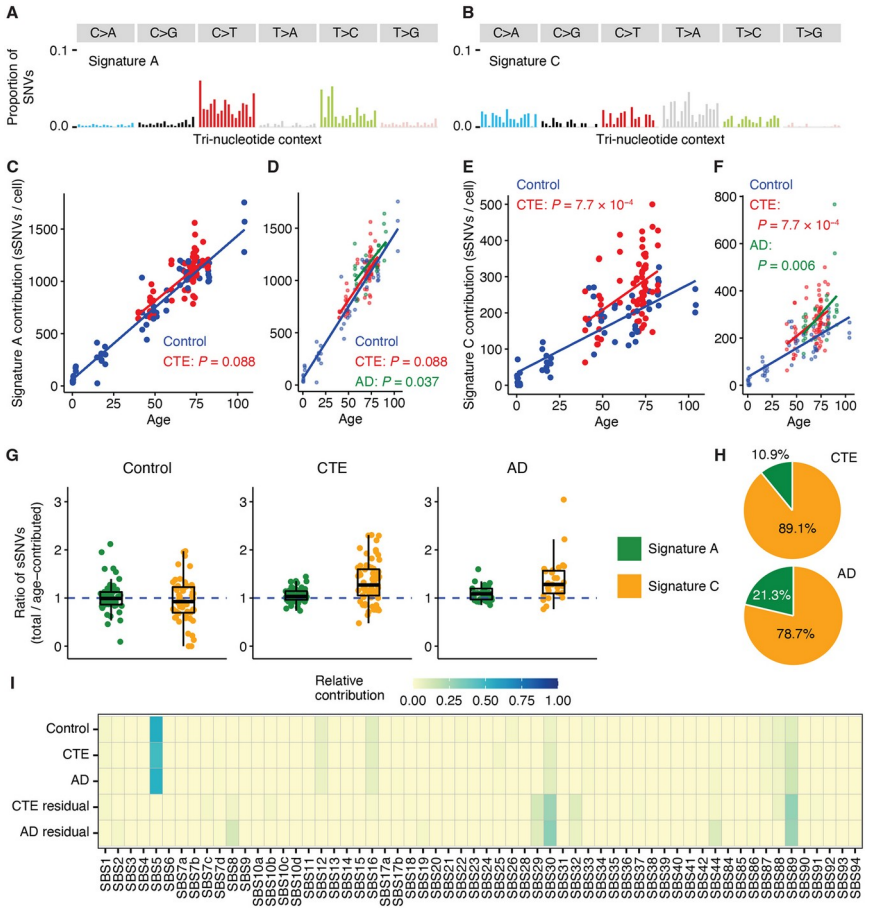
Play video

Oxidative damage can alter a long-lived neuron's genome

Neurons can survive for many decades while maintaining active metabolism. Reactive molecules generated through normal metabolism or cellular stress can damage DNA bases. Repair systems remove or replace damaged material, but persistent or incorrectly repaired changes can alter the sequence of a cell's genome.

A somatic mutation arises within body cells rather than being inherited as the original sequence of every cell in the person. Different neurons can therefore acquire different mutations during life. Finding these changes in brain tissue is not the same as identifying an inherited gene that determines CTE risk.

Researchers can sequence DNA from individual neuronal nuclei to compare the burdens and patterns of change. Age must be considered because mutations also accumulate during ordinary life. A disease-associated excess or a distinctive pattern identifies an additional biological process to investigate beyond normal aging alone.



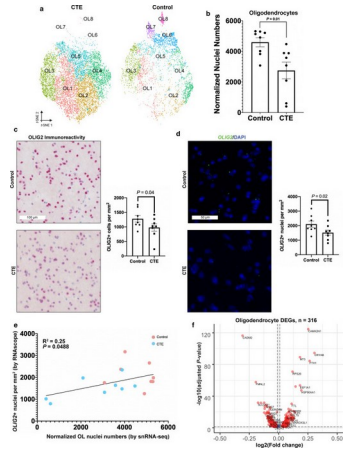
Dong et al. (2025), Fig. 2.

CTE white matter contains more than one altered glial state

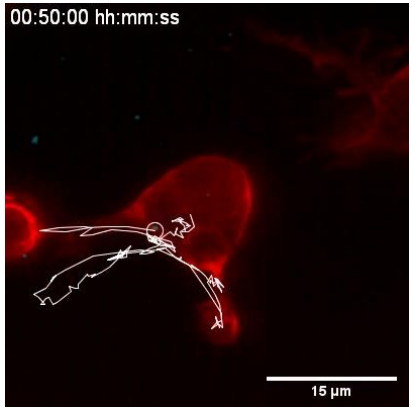
The human white-matter study distinguished reduced oligodendrocyte populations from changes in astrocyte gene expression. Oligodendrocytes maintain the myelin around many axons, while astrocytes support extracellular and metabolic conditions. Their different alterations can influence neural communication through different routes.

Iron-related stress programs and tissue iron accumulation accompanied the oligodendrocyte changes. Astrocytes instead showed stronger inflammatory transcription without the same overall numerical change. The data therefore resist a single description in which every support cell is either lost or uniformly activated.

An independent zebrafish recording follows a microglial centrosome moving into a branch used for engulfment. It demonstrates another glial population's internal machinery rather than oligodendrocyte myelin repair. The comparison separates the molecular states measured in CTE tissue from a directly observed cellular action in a model.



Chancellor et al. (2021), Fig. 2.



Centrosome movement during engulfment. Katrin Möller (2022).

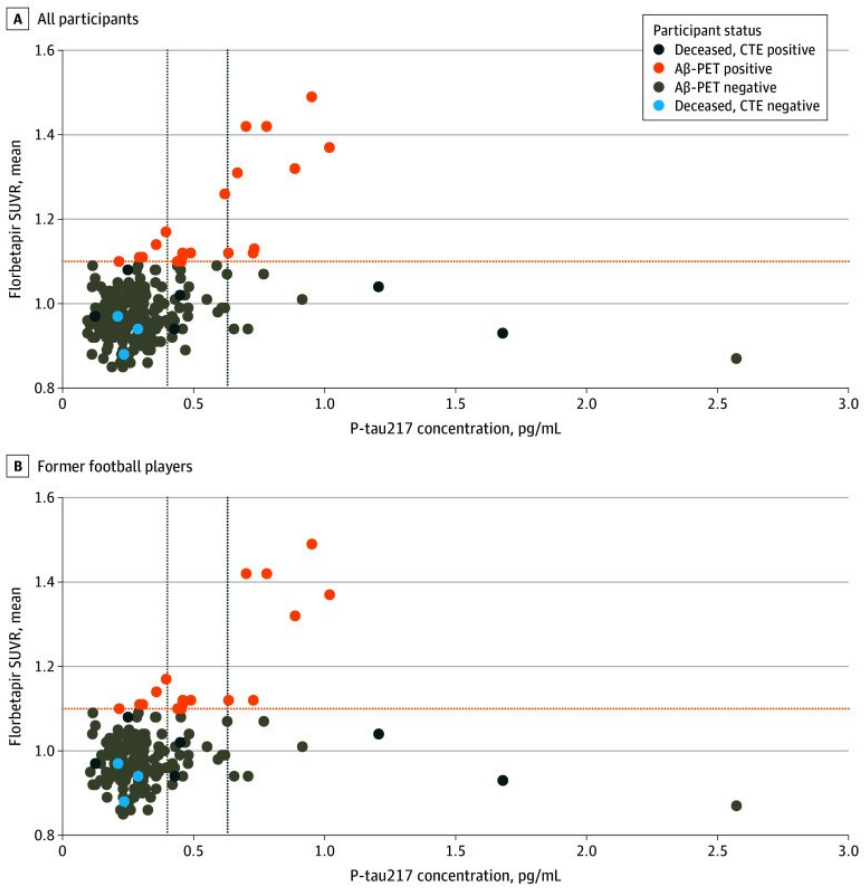
Play video

A biomarker needs a specified biological target

A biomarker is a measurable feature used to indicate a biological process or condition. A blood protein can report neuronal or glial injury without identifying its cause. The intended target must therefore be stated: recent damage, abnormal protein accumulation, a particular disease, or the severity of a functional change.

Neurofilament light is associated with axonal injury, while GFAP reflects an astrocytic structural protein. Their presence in a fluid does not display the perivascular neuronal arrangement required for CTE diagnosis. A general injury signal and a CTE-specific marker therefore answer different questions.

Phosphorylated tau markers also differ by the site they detect and by the diseases in which they perform well. A marker useful for Alzheimer pathology cannot be assumed to detect CTE equally well. Its relationship to the relevant tissue process must be evaluated in the actual population and clinical setting.



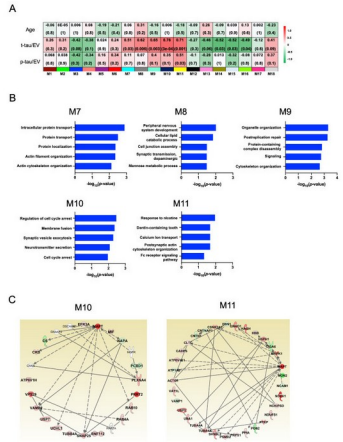
Miner et al. (2026), Fig. 2.

CTE vesicle cargo does not reveal every trafficking step

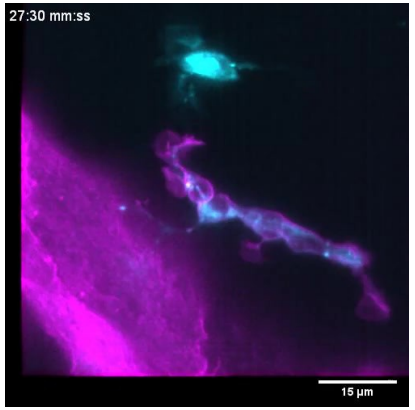
Brain-derived extracellular vesicles from CTE contained altered tau-related cargo and proteins involved in secretion and membrane handling. Their composition supplied evidence about material present outside cells in the affected tissue. It did not identify every intracellular route by which each particle or protein reached that location.

Tau cargo can be examined separately from the machinery that sorts and transports membrane compartments. A vesicle-associated protein may participate in several cellular processes, so its detection does not prove a single spreading mechanism. Establishing transfer requires following movement and testing the receiving cell's response.

The independent microglial recording follows Rab5-associated early endosomes entering a phagocytic branch. Rab5 helps regulate an internal sorting compartment. This is intracellular delivery during engulfment, distinct from demonstrating that a CTE-derived extracellular vesicle released tau and seeded another human neuron.



Muraoka et al. (2021), Fig. 3.



Rab5 endosomes during engulfment. Katrin Möller (2022).

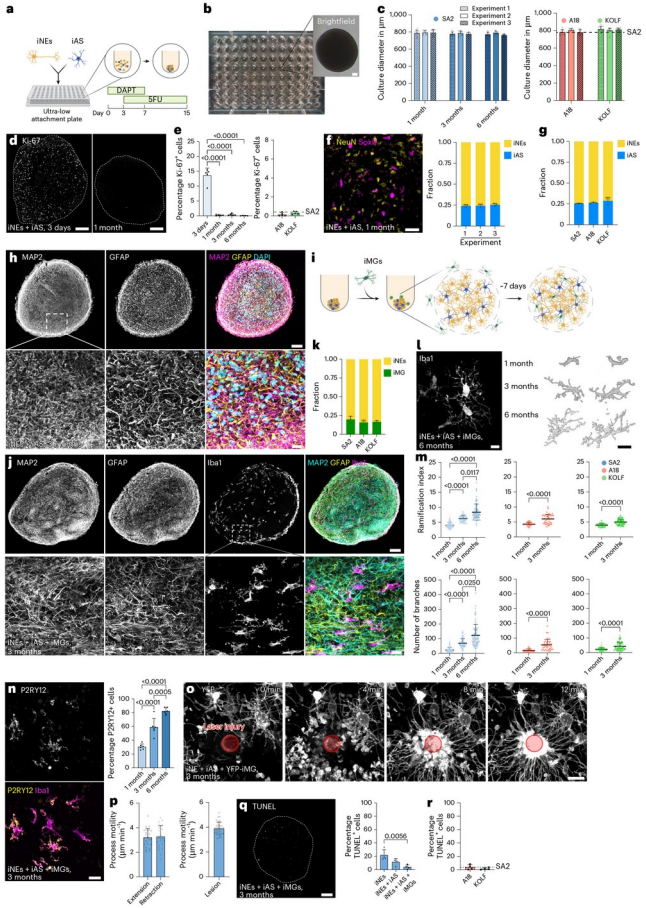
Play video

A CTE model must reproduce a specified disease feature

A model can reproduce an impact, a tau conformer, an inflammatory response, or a particular functional deficit without reproducing every feature of CTE. Its purpose determines the required measurement. An injury apparatus and a cell carrying fluorescent tau therefore test different parts of the disease process.

Human cortical folding, exposure history, aging, and interactions among several cell types influence the setting in which CTE occurs. A simplified preparation may omit one of these features while isolating another. The omitted feature cannot silently be treated as though it had been measured in the model.

Validation compares the experimental outcome with the actual human feature of interest. For a tau-structure question, that may require filament analysis; for a cell-state question, molecular and functional comparisons are needed. A familiar disease label does not substitute for specifying that correspondence.



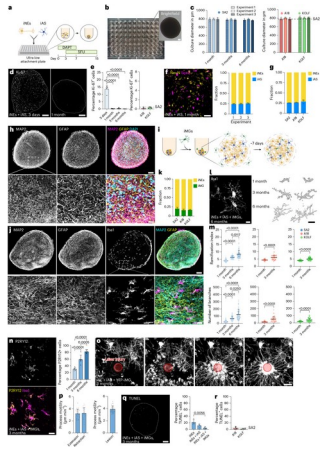
Julien et al. (2026), Fig. 1.

Human brain-tissue models can test a local injury response

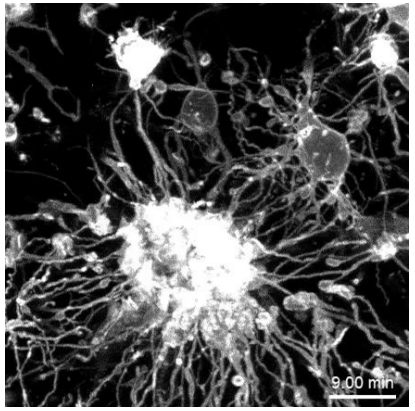
Researchers built a three-dimensional human cortical tissue model from induced pluripotent stem cells. These are cells reprogrammed into a state that can produce several differentiated cell types. The model combined neurons, astrocytes, and microglia rather than studying each population in a separate dish.

The original recording follows microglia responding to focal tissue damage within this preparation. The response provides a functional test of the incorporated immune cells. Their presence can therefore be assessed through an observed action as well as through the expression of microglial markers.

The published platform also modeled Alzheimer-related pathology in separate experiments. The selected injury response is not a demonstration that the system has reproduced CTE. Its experimentally established value is the ability to examine interactions among human brain cell types in a controlled, living tissue environment.



Julien et al. (2026), Fig. 1.



Human 3D microglia respond to tissue damage. Klimmt (2026).

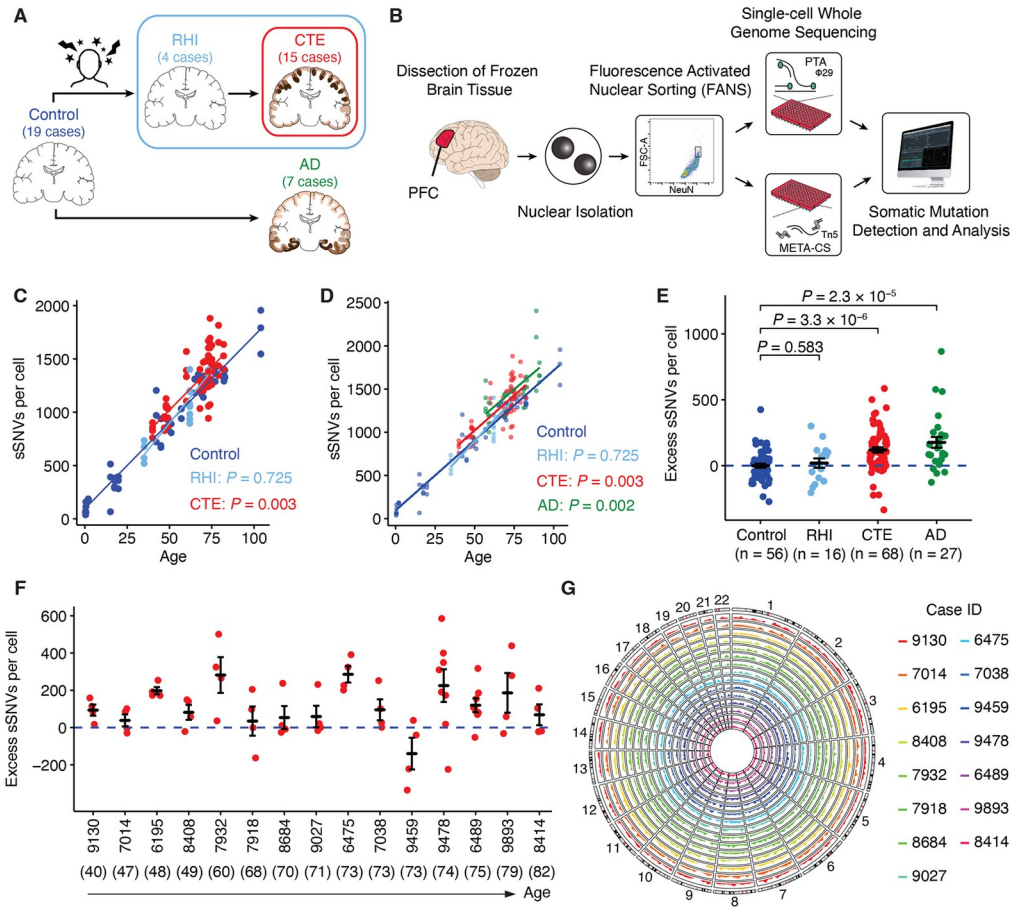
Play video

Individual CTE neurons carry acquired genomic changes

Researchers sequenced whole genomes from individual neuronal nuclei in CTE, repeated-impact, and comparison brains. The approach detected somatic single-base changes and small insertions or deletions. Examining individual cells avoided averaging all neuronal genomes into one apparently uniform tissue sequence.

CTE neurons showed increased somatic mutation burdens with patterns partly resembling those in Alzheimer disease. Some CTE donors also had prominent small insertion and deletion signatures. The repeated-impact group without CTE did not show the same overall pattern of increased burden in the tested comparison.

These results identify acquired genomic damage associated with the disease state, not an inherited football-player gene. They also distinguish repeated-impact exposure from every downstream process found in CTE. The study did not establish whether the mutations initiated degeneration, accumulated during it, or involved both directions over time.



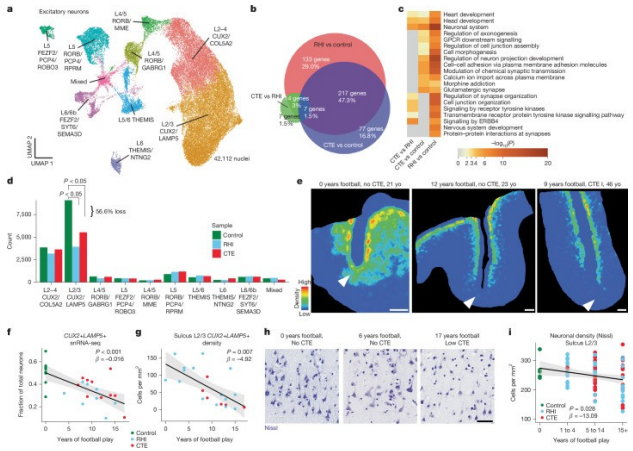
Dong et al. (2025), Fig. 1.

Selective cell loss changes the circuit to be explained

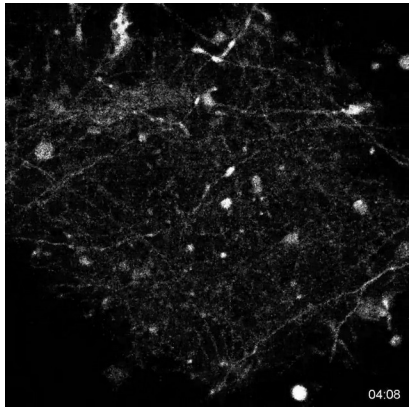
The human cortical study found changes in synaptic gene programs and reductions in particular superficial-layer excitatory populations. Their location matters because cortical layers organize local and long-range communication. Cell identity and anatomical position therefore help specify which circuit interactions may be affected.

RNA measurements establish transcriptional state, while independent staining measures cell density. Neither directly records the surviving network's electrical activity. Functional experiments are needed to test how altered cell composition changes communication, rather than assuming that every molecular difference has the same circuit consequence.

The accompanying human three-dimensional tissue recording follows spontaneous, synchronized calcium activity. It is a separate functional assay in a model, not a measurement from the CTE donors. The example makes visible the network-level outcome that can be investigated alongside human observations of selective cell loss.



Butler et al. (2025), Fig. 4.



Human 3D neural network calcium activity. Klimmt (2026).

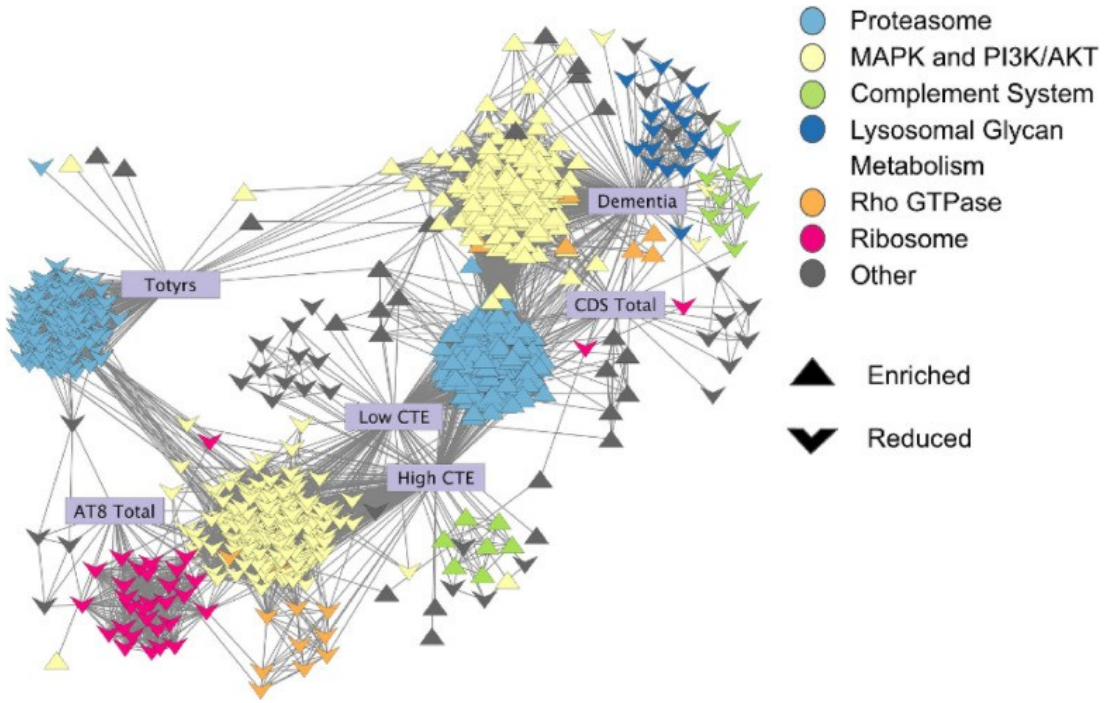
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CTE severity accompanies protein-processing changes

A recent human study measured a broad panel of proteins in dorsolateral prefrontal cortex from repeated-impact and CTE donors. Researchers compared protein patterns with pathological severity, tau burden, exposure duration, and clinical measures. This separated several related features instead of treating them as one disease score.

Proteasome-related pathways were associated with CTE severity and years of contact-sport participation. Ribosomal proteins and pathways were reduced in association with tau burden. Ribosomes assemble proteins from RNA instructions, so the findings concerned both protein production and protein-processing systems.

The study also identified changes in MAPK-related signaling, a family of enzyme pathways that regulate proteins through phosphorylation. Protein abundance did not directly establish the speed of degradation or the function of every ribosome. The results define molecular systems whose activities can be tested in mechanistic experiments.



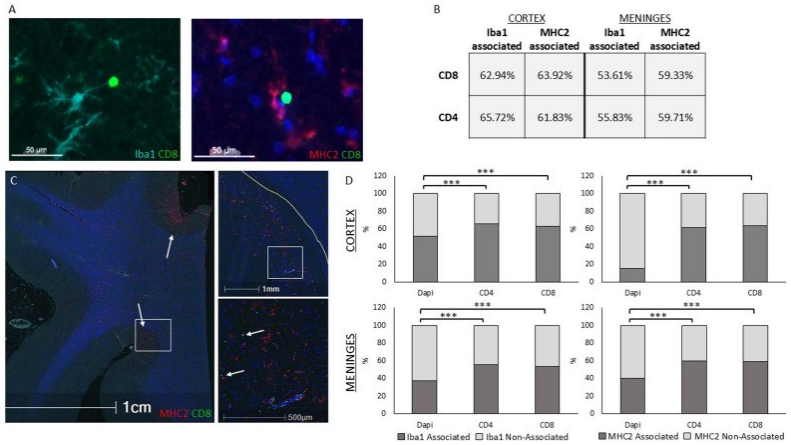
Pennington et al. (2026), Fig. 5.

Synaptic loss differs from direct immune-cell engulfment

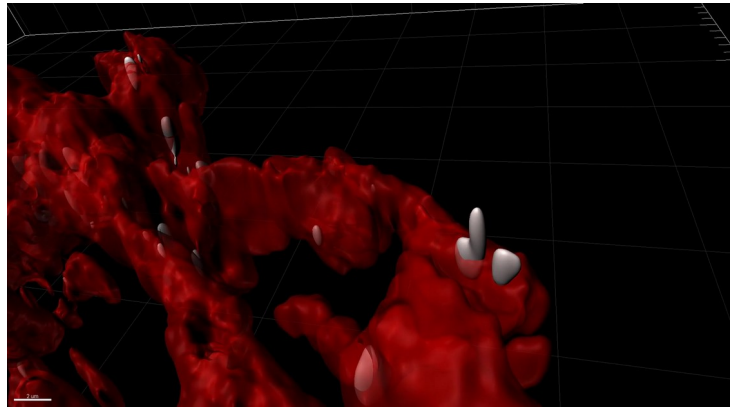
The human T-cell study measured immune populations alongside tau and synaptic markers in anatomically identified regions. Increased infiltration was associated with exposure and synaptic loss, particularly around the sulcal environment. The spatial measurements locate several abnormal processes within a shared tissue neighborhood.

A lower synaptic-marker signal can arise through several biological changes. It does not by itself identify which cell removed a synapse or whether the synapse was already failing. Establishing an engulfment mechanism requires direct evidence of the relevant cell collecting identified synaptic material.

The selected human tissue-model movie follows microglial contact with and engulfment of labeled synaptic material. This is a separate in-vitro preparation, not live imaging of CTE tissue. It demonstrates the cellular action that must be distinguished from the associations measured in the human postmortem study.



Calderazzo et al. (2026), Fig. 3.



Human 3D microglia engulf synaptic material. Klimmt (2026).

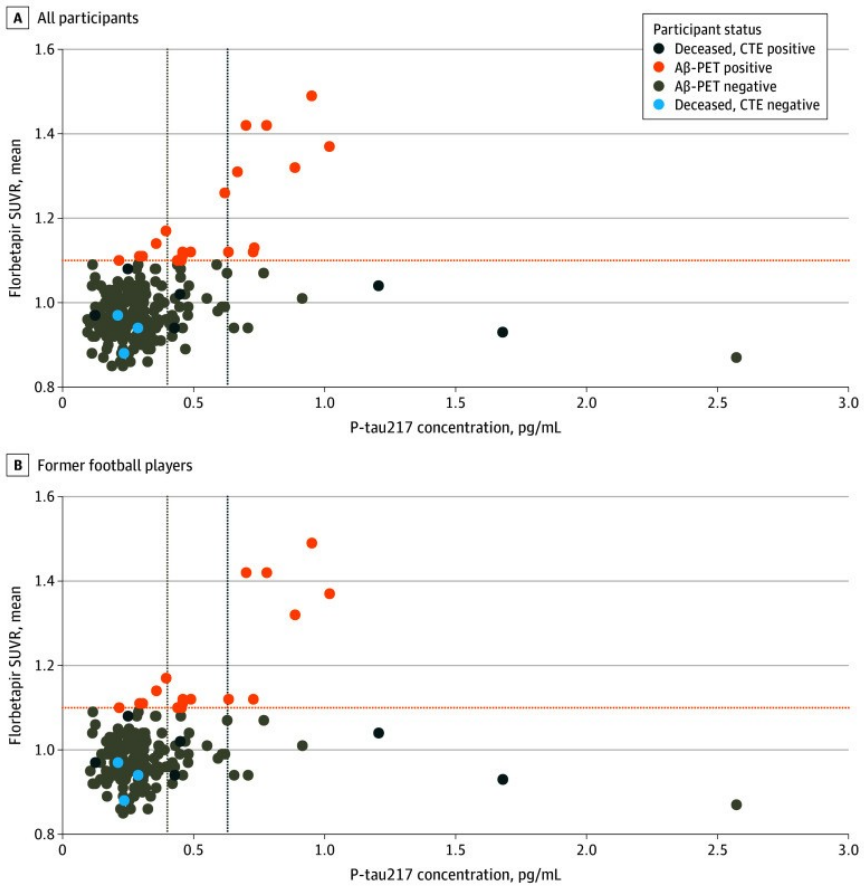
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Plasma p-tau217 mainly identified Alzheimer-related biology

A 2026 study measured plasma tau phosphorylated at position 217 in former football players and unexposed comparison participants. The investigators also examined amyloid imaging and available pathological information. These comparisons tested whether the blood signal identified CTE-related exposure or another neurodegenerative process.

Higher p-tau217 was strongly related to amyloid-positive imaging and Alzheimer disease findings. Changes were not associated with the study's repeated-impact proxies. The results therefore did not support treating this familiar Alzheimer-related marker as a specific indicator of CTE in former players.

A person can have more than one brain pathology, so detecting Alzheimer biology remains clinically informative. It does not determine whether the same person also has CTE pathology. Interpreting the assay requires naming the process it detects rather than converting any abnormal tau-related result into a CTE diagnosis.



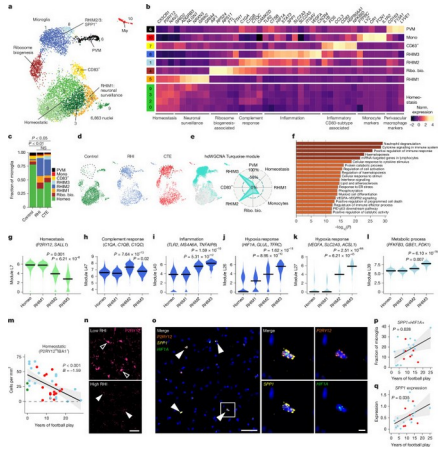
Miner et al. (2026), Fig. 2.

Cell-state markers must be compared with actual behavior

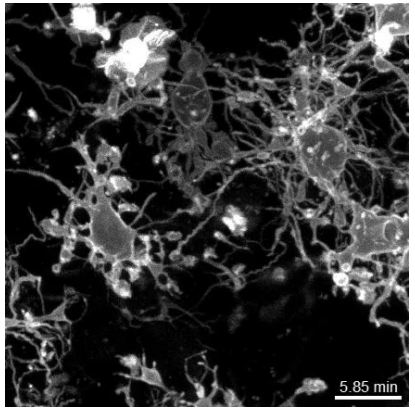
The young-donor study identified inflammatory microglial states and altered interactions with vascular cells after repeated impacts. Such molecular states describe which genes were active in the sampled nuclei. They do not directly show how each living cell surveyed tissue or responded to a damaged synapse.

Functional comparisons require a preparation in which the relevant cells can be observed over time. Cell movement, contact duration, and collection of material are separate actions that can be quantified. The human tissue observations identify which populations and states should be examined in those assays.

The accompanying three-dimensional human model recording follows microglia actively surveying their environment. Its moving processes provide a direct behavioral readout of the cell. The model does not replace the human CTE dataset; it supplies an experimentally accessible function that can be compared with a disease-associated molecular state.



Butler et al. (2025), Fig. 2.



Human 3D microglial surveillance. Klimmt (2026).

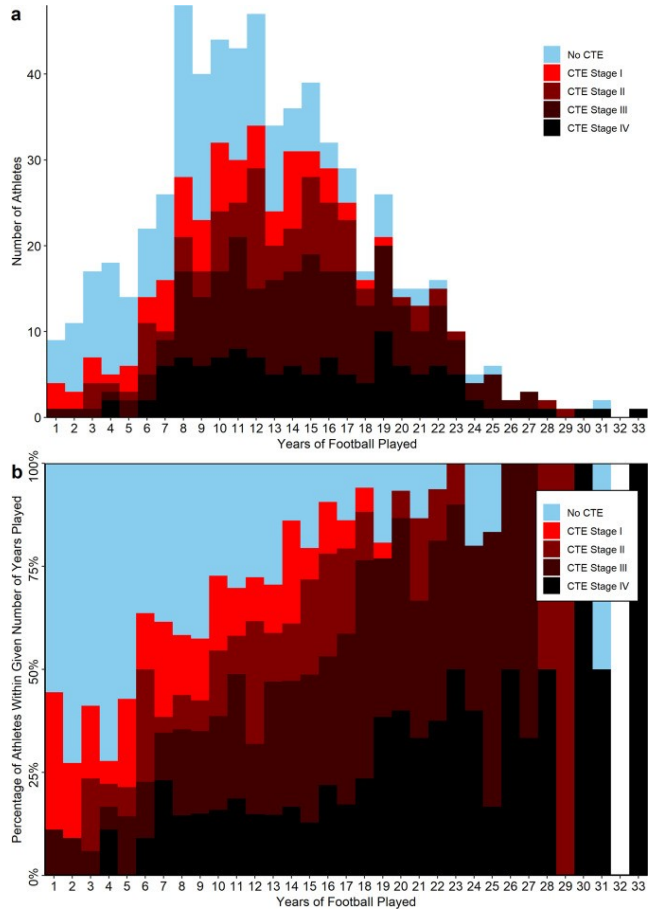
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Prevention targets exposure before a lesion is established

Reducing repeated head-impact exposure acts earlier in the biological sequence than attempting to remove established tau filaments. Rules, practice design, contact opportunities, and appropriate concussion management can alter the circumstances in which injury occurs. Their effects should be measured through actual exposure and health outcomes.

A helmet can reduce some injury risks without eliminating brain acceleration or establishing complete protection from CTE. Acute symptom recovery also does not directly report the long-term fate of tau or vulnerable cell populations. These outcomes occur at different stages and require different measurements.

There is no established disease-modifying cure that reverses confirmed CTE pathology. Care during life addresses the person's actual symptoms and other treatable contributors, while mechanism research tests specific cellular targets. Reducing exposure, managing acute injury, and investigating chronic pathology are distinct interventions within the same biological sequence.



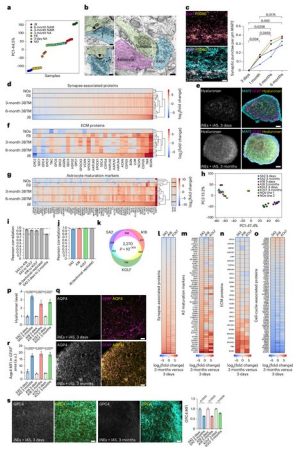
Daneshvar et al. (2023), Fig. 1.

Culture architecture changes the network readout

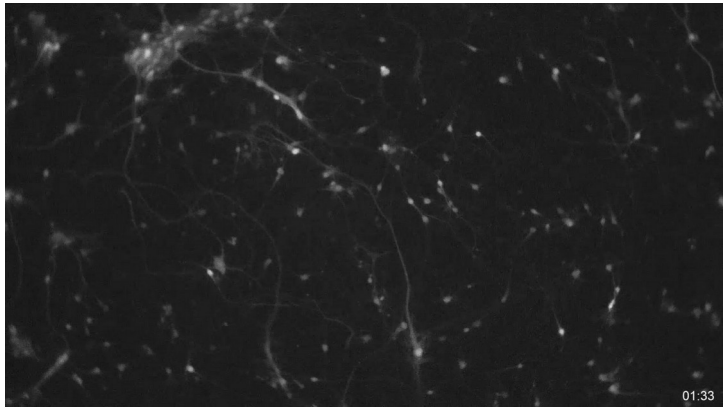
The human model investigators compared three-dimensional tissue with a two-dimensional coculture. Both preparations showed spontaneous calcium activity, but the two-dimensional recording lacked the same network synchronization. A difference in tissue organization therefore changed the functional pattern even before assigning a disease effect.

The original comparison recording follows this two-dimensional activity. The cells are active, so absence of synchrony does not mean that every neuron has stopped functioning. It identifies a particular network property that depends on the preparation and must be considered when interpreting a perturbation.

The three-dimensional platform also maintained incorporated microglia for extended periods and supported interactions among neurons and astrocytes. These demonstrated properties permit controlled tests of multicellular mechanisms. A CTE application would still need disease-relevant tau, exposure conditions, and comparison with the specific human lesion or cell state being investigated.



Julien et al. (2026), Fig. 2.



Human 2D coculture calcium comparison. Klimmt (2026).

Play video

Key takeaways

CTE is a tissue-defined tauopathy. Diagnosis requires the characteristic neuronal phosphorylated-tau lesion around small vessels at cortical sulcal depths; symptoms and photographs cannot establish it in a living player.

Repeated exposure differs from concussion. Football includes diagnosed concussions and repeated impacts without recognized symptoms. Exposure estimates and selected brain donations do not directly establish population prevalence.

Axons and tau have normal functions. Spectrin, actin, microtubules, and tau support axonal mechanics, transport, and synaptic organization. Pathological modification and assembly can disturb these functions.

CTE involves more than tau deposits. Human studies identify selective neuronal loss, white-matter changes, vascular responses, and altered immune-cell states. These measurements test different aspects of neuronal survival and tissue function.

Models answer specified questions. Cell cultures, worms, fish, mice, and human tissue preparations isolate mechanisms. A model of seeding, clearance, or injury is not automatically a model that reproduces CTE.

Clinical evidence must match the claim. Current biomarkers can reflect other pathology, including Alzheimer disease. Exposure reduction, concussion care, and treatment of symptoms remain important while CTE-specific diagnostics and therapies are investigated.